Depth dependence of Gaussian prior costs in critical neural networks

Dmitry Vaintrob Kaarel Hänni OpenAI Codex

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Abstract

For fully connected tanh networks under the critical Gaussian prior, we study the exponential probability cost of realizing a Boolean function with a fixed output margin on the entire input cube. We prove that this cost converges, with increasing depth, to the same universal constant V_* for every Boolean function. The constant satisfies $1 < V_* < 2$ and is characterized by a one-dimensional relative-entropy control problem. Quantitative bounds are uniform over all functions on d inputs. For structured targets, controlled feature laws give sharper bounds: Fourier norm and circuit depth enter through small target components that can be amplified at low entropy cost. In particular, every parity attains a bounded rate at depth $\Theta((\log d)^2)$, the same scale required even by constants and dictators. A layerwise construction also gives polynomial teacher-depth dependence in the rate upper bound for arbitrary bounded-weight, globally margined tanh teachers on the cube.

Every consistently labeled sample of n points has a globally margined extension with rate at most $V_* + O(\log^2(d+n+L)/L)$ at sufficient depth. The actual width-first posterior consequently concentrates below this ceiling. These results do not establish polynomial-depth learning: small differences inside that interval can determine which extensions the posterior prefers. A separate comparison does give a sample lower bound for full parity: at each fixed depth L, $n \le d^{(1-\eta)/(L-1)}$ examples leave limiting Gibbs error $1/2$ with high sample probability, for every fixed $\eta > 0$. This is a polynomial lower bound, not an exponential bound in parity degree. The proofs include the finite-domain large-deviation foundation and the exact feature constructions.

1 Model and meaning of the result

Let $\mathcal{X} = \{-1, 1\}^d$, $d \ge 1$, and $L \ge 2$. A width-N network has $L-1$ hidden layers and computes

$$h^0(x) = \bar{x} = (1, x), \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \quad (1 \le s < L), \qquad z_w(x) = v^T h^{L-1}(x).$$

There are no biases after the augmented input. Under the raw critical Gaussian prior $\mu_{N,1}$, entries are independent centered Gaussians with variances $1/d$ in W^1, $1/N$ in internal matrices, and $1/N^2$ in v. For a Boolean function f, write

$$A_{N,f} = \{w : f(x) z_w(x) > 1/2 \text{ for every } x \in \mathcal{X}\}, \qquad J_L(f) = \lim_{N\to\infty} -\frac{1}{N} \log \mu_{N,1}(A_{N,f}).$$

The limit exists at each fixed d, L. Its proof, including the strict margin convention, appears in Appendix A.

For a feature law ν on $[-1,1]^{\mathcal{X}}$, let $Q(\nu) = \mathbb{E}_\nu hh^T$. Let M_Q be the law of coordinatewise tanh applied to a centered Gaussian with covariance Q, and put $Q_0(x,y) = (1 + x \cdot y)/d$. The rate is

$$J_L(f) = \inf\Big\{ D(\nu_1 \| \mathsf{M}_{Q_0}) + \sum_{s=2}^{L-1} D(\nu_s \| \mathsf{M}_{Q(\nu_{s-1})}) + \tfrac{1}{2} z^T Q(\nu_{L-1})^+ z : \\ f(x) z(x) \ge 1/2 \quad (x \in \mathcal{X}) \Big\}.$$

The last term is infinite outside the covariance range. The matrices $Q(\nu)$ are uncentered second moments, not necessarily centered covariances.

The main result, Theorem 2.1, identifies a universal constant V_* such that, with $D = d + 1$,

$$V_* - \frac{C}{L-1} \le J_L(f) \le V_* + C\frac{D^2 + [\log(2L)]^2}{L} \qquad (L \ge CD).$$

The same bound holds for every Boolean function at the same specified depth. In particular, an absolute rate budget cannot provide a polynomial bound on the logarithm of the number of possible functions: at sufficiently large polynomial depth, a fixed budget already contains all 2^{2^d} functions.

There are sharper upper bounds for small implementations. Corollary 15.2 applies to any depth-D, width-C tanh teacher with weights bounded by B and signed output at least $1/2$ on the cube. Put $R = d + C + B + D + L + 2$. At learner depth $L \ge A(D+1)\log R$, it gives

$$J_L(f) \le V_* + A\frac{(D+1)^2 \log^2 R}{L}.$$

The constant A is universal. Thus exponential teacher-depth dependence can be removed from this critical prior-rate upper bound without restricting the prior's support. The proof compiles the teacher at finite precision, propagates exact Boolean gate vectors through low-entropy mixture laws, and then amplifies the target component. Theorem 12.1 supplies another bound through the Fourier ℓ^1 norm. In particular, all parities attain a bounded rate at depth of order $(\log d)^2$.

Theorem 16.1 gives a different, sample-dependent comparison: every consistent sample of at most n points has an extension with rate at most $V_* + A\log^2(d+n+L+2)/L$ when $L \ge A\log(d+n+L+2)$. The actual width-first posterior concentrates below this ceiling. This narrows the interval in which accurate and inaccurate extensions must be compared; it does not supply that comparison.

This is not a claim that the conditioned prior is approximately uniform. For a fixed training sample S, its exponential posterior rate at a consistent function is

$$J_L(f) - \min_{g \text{ consistent with } S} J_L(g).$$

Any positive difference, however small, is multiplied by width N. The theorem neither orders these differences nor controls finite-width probabilities to accuracy $1/N$. The general sample-complexity question for small deep circuits remains open. All limits in width are taken with dimension and depth fixed before studying the resulting function J_L as depth varies.

A complementary theorem describes the cost scale at every depth. Put

$$\Phi_L(d) = 1 + L\big(d^{1/(2L)} - 1\big)^2.$$

Theorem 22.1 gives universal constants $0 < c < C$ such that every target has $J_L(f) \ge c\Phi_L(d)$, while constants and dictators have $J_L(f) \le C\Phi_L(d)$. Thus this expression characterizes the minimum absolute-rate scale without asserting which function minimizes the exact rate at every dimension and depth. Later sections give a target-relative separation and a polynomial sample lower bound at fixed depth, with an extension to a specified slowly growing-depth regime.

The proof first supplies a finite-width exponential potential. It then uses a quantitative first-layer spectrum bound to seed an arbitrary Boolean component, reduces the unused variance without entropy cost, and amplifies that component through small covariance changes. A scalar control problem describes the universal final cost. The arguments have received independent AI-agent mathematical review; they have not been formalized in Lean. Constants are not optimized.

2 Main theorem

The limiting constant is the scalar entropy-control value V_* from Definition 7.1. Its definition involves only one-dimensional Gaussian laws and the activation tanh.

Theorem 2.1 (A common critical rate at large depth)**.** *There are universal constants $C < \infty$ and $V_* \in (1, 2)$ such that, for every input dimension $d \geq 1$, depth $L \geq 2$, and Boolean function $f : \{-1, 1\}^d \to \{-1, 1\}$,*

$$J_L(f) \geq V_* - \frac{C}{L-1}.$$

With $D = d + 1$, whenever $L \geq CD$, one also has

$$J_L(f) \leq V_* + C\frac{D^2 + [\log(2L)]^2}{L}.$$

Both bounds hold uniformly over all Boolean functions at the same specified depth. In particular, at every fixed input dimension, $J_L(f) \to V_*$ uniformly in f as $L \to \infty$.

Proof. Lemma 7.2 constructs V_* and proves $1 < V_* < 2$. The lower estimate is proved in Theorem 9.2; the upper estimate is proved in Theorem 11.1. Increasing the common universal constant gives the displayed simultaneous bounds. □

For every fixed $\varepsilon > 0$, a depth polynomial in d and $1/\varepsilon$ therefore places all 2^{2^d} Boolean functions inside the same absolute rate interval $[V_* - \varepsilon, V_* + \varepsilon]$. An absolute rate sublevel cannot by itself provide a polynomial-depth bound on the logarithm of the number of possible functions. This statement does not identify the ordering of rates inside the interval; such remaining differences can still determine a width-first posterior.

3 A depth-independent lower bound on every critical rate

At every prior scale $0 < t \leq 1$, the global clean-margin event has an exponentially small probability, uniformly in depth. The exponent has a universal positive lower bound as the input dimension grows. This result does not compare the excess rates of different Boolean functions and is not a posterior sample-complexity lower bound.

Throughout this section the hidden activation is tanh, the input is $\bar{x} = (1, x)$ for $x \in \{-1, 1\}^d$, the first-layer variance is t^2/d, the subsequent hidden-weight variance is t^2/N, and the readout variance is t^2/N^2, where $0 < t \leq 1$. There are $L - 1$ hidden layers, with $L \geq 2$. Write $A_{N,f}$ for the event $f(x)z(x) > 1/2$ at every cube input, and $A_N = \bigcup_f A_{N,f}$ for the global clean-margin event.

Lemma 3.1 (A scalar exponential potential)**.** *For $0 < t \leq 1$, $0 \leq a \leq 1$, and $G \sim \mathcal{N}(0, t^2 q)$, $q \geq 0$,*

$$\mathbb{E} \exp\{(2a/t^2) \tanh^2 G\} \leq e^{2aq/t^2}.$$

If (G_x, G_y) is any centered Gaussian pair, $s \in \{-1, 1\}$, and $t^2 v = \mathrm{Var}[(G_x + sG_y)/2]$, then

$$\mathbb{E} \exp\left\{\frac{2a}{t^2}\left(\frac{\tanh G_x + s \tanh G_y}{2}\right)^2\right\} \leq e^{2av/t^2}.$$

Proof. The case $a = 0$ is immediate. Otherwise put $\lambda = 2a/t^2$ and $F_\lambda(z) = \exp(\lambda \tanh^2 z)$. Writing $u = \tanh^2 z \in [0, 1)$, direct differentiation gives

$$\frac{F_\lambda''(z)}{F_\lambda(z)} = 2\lambda(1-u)[1 - 3u + 2\lambda u(1-u)] \le \frac{2\lambda}{t^2}.$$

To check the inequality, multiply its bracket by t^2. The resulting quantity $t^2(1-u)(1-3u)+4au(1-u)^2$ is affine in $t^2 \in [0, 1]$. At the two endpoints it is at most $16/27$ and $1 - 5u^2 + 4u^3 \le 1$, respectively. The Gaussian heat equation and Gronwall's inequality therefore give $\mathbb{E}F_\lambda(\sqrt{r}Z) \le \exp(\lambda r/t^2)$ for $r \ge 0$. All derivatives needed here are bounded for each fixed λ. Taking $r = t^2 q$ proves the first assertion. The elementary identities for the sum and difference of two hyperbolic tangents imply

$$\left|\frac{\tanh a + s \tanh b}{2}\right| \le \tanh\left|\frac{a + sb}{2}\right|.$$

For example, after writing $a = A + D$, $b = A - D$, the half-sum equals $\sinh(2A)/[\cosh(2A) + \cosh(2D)]$, whose absolute value is at most $\tanh|A|$. The half-difference is identical with A, D interchanged. Applying the scalar assertion to $(G_x + sG_y)/2$ proves the claim, without any nonsingularity assumption. □

Theorem 3.2 (Finite-width clean-margin upper bounds for all $\gamma \ge 1$)**.** *For every $d \ge 1$, $N \ge 1$, $L \ge 2$, $0 < t \le 1$, and every Boolean function f,*

$$\mu_{N,t}(A_{N,f}) \le \exp\{-(N/t^2)\psi(d)\}, \qquad \psi(d) = \begin{cases} d/8, & d \le 4, \\ 1 - 2/d, & d \ge 4. \end{cases}$$

The unconditioned Gaussian probability of global clean margin satisfies

$$\mu_{N,t}(A_N) \le \min\{1,\ 2^{d+1} \exp[-(N/t^2)\psi(d)]\}.$$

In the usual scaling $t^2 = N^{1-\gamma}$, the exponential speed N/t^2 is N^γ, and the assumption is $\gamma \ge 1$. The first bound holds with no factor depending on depth or width outside its exponential. The simpler exponent $1 - 2/d$ is valid in every dimension.

Proof. Fix two inputs x, y, a choice $s \in \{-1, 1\}$, and $a \in (0, 1]$. For the empirical hidden feature covariance Q_j, set

$$v_j = \frac{1}{4}(Q_j(x,x) + Q_j(y,y) + 2sQ_j(x,y)), \qquad 0 \le j \le L-1,$$

where $Q_0(x, y) = \bar{x}\cdot\bar{y}/d$. Conditional on the previous layer, the N preactivation pairs are independent centered Gaussians with covariance given by $t^2 Q_{j-1}$. Lemma 3.1 therefore gives

$$\mathbb{E}[\exp((2aN/t^2)v_j) \mid \text{previous layers}] \le \exp((2aN/t^2)v_{j-1}).$$

Consequently $\mathbb{E}\exp((2aN/t^2)v_{L-1}) \le \exp((2aN/t^2)v_0)$.

Conditional on the hidden layers, the readout combination $Z = (z(x) + sz(y))/2$ is centered Gaussian with variance $t^2 v_{L-1}/N$. For either specified sign of Z, its probability of exceeding $1/2$ in that sign is at most $\exp[-N/(8t^2 v_{L-1})]$, interpreted as zero when $v_{L-1} = 0$. Since

$$\frac{1}{8v} \ge \sqrt{a} - 2av \qquad (v > 0),$$

the unconditional one-sided probability is at most $\exp[-(N/t^2)(\sqrt{a}-2av_0)]$.

If f is nonconstant, some adjacent cube vertices have opposite labels. Choose such x, y and $s=-1$. Their input difference has squared norm four, so $v_0 = 1/d$, and $A_{N,f}$ implies a specified sign of Z exceeds $1/2$. If f is constant, use antipodal inputs $y=-x$ and $s=+1$. Their augmented-input half-sum is the bias coordinate, again giving $v_0 = 1/d$. Maximizing $\sqrt{a}-2a/d$ over $a \in [0,1]$ gives $\psi(d)$, proving the first assertion.

For the second assertion fix a spanning tree of the cube graph and one antipodal pair. Every nonconstant sign function changes sign on one of the tree's 2^d-1 edges; a constant function has equal signs on the fixed antipodal pair. Each case implies $|Z| > 1/2$ for its corresponding difference or sum. The two-sided bound and a union bound over these 2^d pairs give the result. □

Corollary 3.3 (Universal lower critical-rate baseline)**.** *For the critical variational rate functional J_L with output margin $1/2$, every Boolean function satisfies*

$$J_L(f) \geq \psi(d) \geq 1 - \frac{2}{d} \qquad (L \geq 2).$$

Proof. This can be read directly from the finite-width upper bound whenever the corresponding large-deviation formula is available. There is also a direct proof from the variational functional, which avoids any interchange of width and depth limits. Choose the signed pair from the preceding proof, put $b = (e_x + s e_y)/2$, and write $v_j = b^T Q_j b$ for the covariances along any admissible variational path. For any $a \in (0,1]$, the entropy variational inequality and Lemma 3.1 at $t=1$ give, for every hidden layer,

$$D(\nu_j \| \mathsf{M}_{Q_{j-1}}) \geq 2a(v_j - v_{j-1}).$$

The output constraints imply $|b^T z| \geq 1/2$. If the readout cost is finite, Cauchy–Schwarz on the range of Q_{L-1} gives

$$\tfrac{1}{2} z^T Q_{L-1}^{+} z \geq \frac{1}{8 v_{L-1}}.$$

The inequality is immediate if this cost is infinite. Summing the entropy inequalities and minimizing over $v = v_{L-1} > 0$ now yields

$$J_L(f) \geq \inf_{v>0} \left(2av + \frac{1}{8v}\right) - 2av_0 = \sqrt{a} - \frac{2a}{d}.$$

Maximizing in a gives $\psi(d)$. □

For completeness, the same argument with an arbitrary required output margin $r > 0$ replaces $\psi(d)$ by

$$\psi_r(d) = \max_{0 \leq u \leq 1} (2ru - 2u^2/d) = \begin{cases} r^2 d/2, & rd \leq 2, \\ 2r - 2/d, & rd \geq 2. \end{cases}$$

This applies to both finite-width bounds and the corresponding critical variational rate. No change to the network architecture is needed.

The lower bound supplies a common baseline for rate comparisons. It does not identify the minimum rate, bound the difference between a target's rate and a competitor's rate, or show that the posterior is diffuse among functions of bounded rate. In particular, even a small positive rate gap is amplified by width in the critical posterior.

4 A quantitative first-layer spectrum bound

The proof starts with a quantitative lower bound on every Fourier eigenvalue of the first tanh feature kernel.

Lemma 4.1 (A quantitative first-layer spectrum bound)**.** *Let $d \geq 1$, $D = d + 1$, $m = 2^d$, and let Q be the $m \times m$ matrix*

$$Q(x, y) = \mathbb{E}_w \tanh(w \cdot \bar{x}) \tanh(w \cdot \bar{y}), \qquad w \sim \mathcal{N}(0, I_{d+1}/d).$$

Then

$$\lambda_{\min}(Q) \geq e^{-13D}.$$

Proof. Write $q = D/d \in [1, 2]$, and let h_n be the coefficient of the normalized probabilists' Hermite polynomial $\mathrm{He}_n / \sqrt{n!}$ in $z \mapsto \tanh(\sqrt{q}z)$. The coefficients vanish for even n. For every odd $n \geq 1$, the Fourier sine identity

$$\tanh(\sqrt{q}z) = \frac{1}{\sqrt{q}} \int_0^\infty \frac{\sin(tz)}{\sinh(\pi t/(2\sqrt{q}))}\, dt$$

and $\mathbb{E}[\mathrm{He}_n(Z) e^{itZ}] = (it)^n e^{-t^2/2}$ give

$$|h_n| = \frac{1}{\sqrt{qn!}} \int_0^\infty \frac{t^n e^{-t^2/2}}{\sinh(\pi t/(2\sqrt{q}))}\, dt.$$

The sine identity follows, for example, by expanding $1/\sinh(\pi t/2) = 2\sum_{j \geq 0} e^{-\pi(j+1/2)t}$ and using the logarithmic derivative of the Euler product for cosh. Fubini is valid in the displayed Hermite integral: near zero use $|\sin(tZ)| \leq t|Z|$, and at infinity the reciprocal hyperbolic sine has exponential decay.

Restrict the positive integral to $[\sqrt{n}, \sqrt{n} + 1]$. Using $n! \leq n^n$, $1/\sinh u \geq 2e^{-u}$, and $1 \leq q \leq 2$ gives

$$|h_n| \geq \sqrt{2} \exp\left[-\frac{n}{2} - \left(1 + \frac{\pi}{2}\right)\sqrt{n} - \frac{1 + \pi}{2}\right] \geq e^{-6n}.$$

The last inequality follows from $\sqrt{n} \leq n$, $n \geq 1$ and $\pi < 4$.

Mehler's identity now yields the nonnegative power series

$$Q(x, y) = \sum_{n \text{ odd}} h_n^2 \left(\frac{1 + \sum_{i=1}^d x_i y_i}{D}\right)^n.$$

The Boolean characters diagonalize this translation-invariant kernel. For a character of degree r, choose the odd integer $n = r$ if r is odd and $n = r + 1$ if r is even. Then $1 \leq n \leq D$. In the multinomial expansion of the nth power, choosing every selected coordinate exactly once, and choosing the constant coordinate once when r is even, contributes $n!/D^n$ to this character's normalized eigenvalue. Every other contribution is nonnegative. Therefore its eigenvalue for Q/m is at least

$$h_n^2 \frac{n!}{D^n} \geq e^{-12n - D} \geq e^{-13D}.$$

Here $n! \geq (n/e)^n$, and $n[1 + \log(D/n)] \leq D$, prove $n!/D^n \geq e^{-D}$. Multiplying by $m \geq 1$ proves the stated lower bound for Q. □

5 A subexponential lower bound for the first-layer spectrum

Theorem 5.1 (Subexponential spectral floor)**.** *Let $D = d + 1 \geq 2$, $m = 2^d$, and*

$$K(x, y) = \mathbb{E}_w \tanh(w \cdot \bar{x}) \tanh(w \cdot \bar{y}), \qquad w \sim \mathcal{N}(0, I_D/d).$$

Then

$$\lambda_{\min}(K) \geq p_d := \exp\{-13\sqrt{D \log(4D)}\}.$$

Proof. Set $q = D/d \in [1, 2]$. The positive integral formula from Lemma 4.1 gives, for every odd $n \geq 1$,

$$|h_n| = \frac{1}{\sqrt{qn!}} \int_0^\infty \frac{t^n e^{-t^2/2}}{\sinh(\pi t/(2\sqrt{q}))}\, dt,$$

where h_n is the normalized Hermite coefficient of $z \mapsto \tanh(\sqrt{q}z)$. Use the elementary Stirling upper bound $n! \leq 3\sqrt{n}(n/e)^n$ and restrict the integral to $[\sqrt{n}, \sqrt{n} + 1]$. Since $1/\sinh u \geq 2e^{-u}$,

$$|h_n| \geq \sqrt{2/3}\, n^{-1/4} \exp\left[-(1 + \pi/2)\sqrt{n} - (1 + \pi)/2\right] \geq e^{-6\sqrt{n}}.$$

For the last inequality, use $\pi < 4$, $\log n \leq \sqrt{n}$, and $\sqrt{n} \geq 1$.

The normalized Walsh coefficient of

$$\left(\frac{1 + \sum_{i=1}^d z_i}{D}\right)^n$$

at a subset $S \subseteq \{1, \ldots, d\}$ is the probability $P_n(S)$ that the lazy walk on $\mathbb{F}_2^d$, choosing uniformly between staying put and toggling any of its d coordinates, reaches S after n steps. Fourier inversion gives

$$|mP_n(S) - 1| \leq \sum_{j=1}^d \binom{d}{j} \left|1 - \frac{2j}{D}\right|^n.$$

For $1 \leq j \leq D - 1$, put $k = \min(j, D - j)$ and bound the last factor by $e^{-2nk/D}$. Pairing $j = k$ with $j = D - k$ and using Pascal's identity bounds the sum by

$$\sum_{k=1}^{\lfloor D/2 \rfloor} \binom{D}{k} e^{-2nk/D} \leq (1 + e^{-2n/D})^D - 1.$$

At the middle index when D is even this only overcounts, so the inequality remains valid. Choose the smallest odd integer $n \geq (D/2)\log(4D)$. Then $n \leq D\log(4D)$ and the displayed sum is at most $e^{1/4} - 1 < 1/2$. Thus $mP_n(S) \geq 1/2$ for every S.

The kernel's Hermite expansion has nonnegative coefficients, so every eigenvalue of K is at least

$$mh_n^2 P_n(S) \geq \frac{1}{2} e^{-12\sqrt{n}} \geq e^{-13\sqrt{D\log(4D)}}.$$

□

Corollary 5.2 (Empirical first-layer kernel)**.** *Let $h_1, \ldots, h_N$ be independent first-layer feature vectors and $\widehat{K} = N^{-1} \sum_i h_i h_i^T$. For $0 < \delta < 1$, if*

$$N \geq \frac{8m}{p_d} \log \frac{m}{\delta},$$

then $\widehat{K} \succeq K/2 \succeq (p_d/2) I_m$ with probability at least $1 - \delta$.

Proof. The whitened summands $K^{-1/2} h_i h_i^T K^{-1/2}$ have expectation I_m and operator norm at most m/p_d, because $\|h_i\|_2^2 \leq m$. The lower-tail matrix Chernoff inequality gives failure probability at most $m \exp[-Np_d/(8m)]$. □

6 Free critical cooling preserves a small target component

Write $F(q) = \mathbb{E}\tanh^2(\sqrt{q}Z)$ for a standard Gaussian Z. For a feature second moment Q, let M_Q denote the law of coordinatewise tanh of a centered Gaussian with covariance Q.

Lemma 6.1 (Uniform free-cooling estimates)**.** *There are universal constants $B < \infty$ and $b > 0$ with the following property. Suppose Q_0 has constant diagonal $q_0 \in [F(1), F(2)]$ and $Q_0 \succeq v_0 f f^T$, where f is Boolean and $v_0 > 0$. Successively apply the unmodified feature law M_Q. For every $0 < \delta \le 1/128$, the first layer whose diagonal q_k is at most δ occurs after at most B/δ transitions. At this layer,*

$$\delta/2 \le q_k \le \delta, \qquad Q_k \succeq b v_0 \delta \, f f^T.$$

All these transitions cost zero in the critical variational functional.

Proof. The diagonal evolves exactly by $q_{j+1} = F(q_j)$. The function F is increasing and $0 < F(q) < q$ for $q > 0$. Taylor expansion with bounded derivatives gives

$$F(q) = q - 2q^2 + O(q^3).$$

It follows by continuity on $(0, F(2)]$ and the limit at zero that there is a universal $c_F > 0$ satisfying $q - F(q) \ge c_F q^2$ throughout this interval. Hence $1/q_{j+1} - 1/q_j \ge c_F$, which bounds the first hitting time of $(0, \delta]$ by $O(1/\delta)$. The elementary bound $F(q) \ge q/(1+3q)$ follows from $\tanh^2 t \ge t^2/(1+t^2)$ and Cauchy–Schwarz. Monotonicity therefore implies that the first value below δ is at least $F(\delta) \ge \delta/(1+3\delta) \ge \delta/2$.

If $Q \succeq v f f^T$ has constant diagonal q, decompose its Gaussian as $\sqrt{v}\, f Z + G$, with G independent and covariance $Q - v f f^T$. The conditional feature mean is $f\psi(Z)$, with the same scalar function at every input because all residual variances equal $q - v$. Conditional expectation, projection onto the first Hermite polynomial, and Gaussian integration by parts give

$$Q' \succeq \mathbb{E}\psi(Z)^2 f f^T \succeq v[1 - F(q)]^2 f f^T.$$

Thus we may track $v_{j+1} = v_j[1 - F(q_j)]^2$. The ratio v_j/q_j is multiplied by

$$R(q_j) = \frac{q_j[1 - F(q_j)]^2}{F(q_j)}.$$

The expansion of F yields $R(q) = 1 + O(q^2)$ at zero, and R is positive on $(0, F(2)]$. Consequently $|\log R(q)| \le C_R q^2$ there for a universal C_R. Also

$$\sum_{j<k} q_j^2 \le c_F^{-1} \sum_{j<k} (q_j - q_{j+1}) \le F(2)/c_F.$$

The product of all ratio factors is therefore at least a universal positive constant. Since $q_0 < 1$ and $q_k \ge \delta/2$, this gives $v_k \ge b v_0 \delta$ for a universal $b > 0$. □

7 Scalar control and target-component constructions

We establish the scalar value and the constructions used in the proof of Theorem 2.1. The critical variational functional is the one defined in the model section, with output margin $1/2$. All logarithms are natural.

Definition 7.1 (Scalar control value)**.** *For $q > 0$, a scalar path of length $m \geq 0$ starts at $q_0 = q$. At step j, choose a probability law η_j on $\mathbb{R}$ with finite entropy relative to $\mathcal{N}(0, q_{j-1})$, and set*

$$q_j = \int \tanh^2 z \, d\eta_j(z).$$

Its total cost is

$$\sum_{j=1}^{m} D(\eta_j \| \mathcal{N}(0, q_{j-1})) + \frac{1}{8q_m}.$$

Let $V(q)$ be the infimum of these costs over all finite paths, including the length-zero path. Define

$$V_* := \lim_{q \downarrow 0} V(q).$$

The next lemma establishes that this limit exists and is finite.

7.1 Properties of the scalar value

Lemma 7.2 (Scalar value and increasing controls)**.** *The function V is nonincreasing, $1 < V_* < 2$, and there is a universal constant C_0 such that*

$$0 \leq V_* - V(q) \leq C_0 q \qquad (0 < q \leq 1/128).$$

For every $0 < p \leq 1/128$, a scalar path of cost at most $V(p) + p$ can be chosen with strictly increasing states.

Proof. Monotonicity follows from rescaling controls. If one starts a path at a larger variance, rescale its first preactivation control by the square root of the variance ratio. Relative entropy from the corresponding Gaussian is unchanged, while the next second moment increases because $|\tanh z|$ increases with $|z|$. Repeat this at every step. The final readout cost decreases. Taking infima proves that V is nonincreasing.

Put $F(u) = \mathbb{E} \tanh^2(\sqrt{u} Z)$. The elementary inequality $\tanh^2 z \geq z^2/(1 + z^2)$ and Cauchy–Schwarz give $F(u) \geq u/(1 + 3u)$. Here is an elementary uniform upper bound. For $q < 1/10$, replace its reference Gaussian by $\mathcal{N}(0, (1 + 6q)q)$, at entropy cost at most $9q^2$. The bound $F(u) \geq u/(1 + 3u)$ gives

$$q' - q \geq \frac{q^2(3 - 18q)}{1 + 3q + 18q^2} > 0.$$

The cost divided by this variance increment is at most

$$R_0(q) = \frac{3(1 + 3q + 18q^2)}{1 - 6q},$$

which is increasing on $[0, 1/10]$. Repeatedly take full steps while the new state remains below $1/10$. Their total cost is bounded by

$$\int_0^{1/10} R_0(q) \, dq = -\frac{69}{200} + \log(5/2).$$

On the last step, continuously reduce the variance inflation to hit $1/10$ exactly; this costs at most $9/100$. The readout then costs $5/4$. The total is at most $199/200 + \log(5/2) < 2$. For starting states at least $1/10$, the length-zero path already costs at most $5/4$. Thus $V(q) < 2$ for all $q > 0$. Since V is nonincreasing, its limit at zero exists and is at most two.

For a quantitative small-state bridge, let $q \le p_0 := 1/128$ and replace its reference Gaussian by $\mathcal{N}(0,(1+16q)q)$. The entropy cost is at most $64q^2$. The first Hermite coefficient and Gaussian integration by parts give

$$F(u) \ge u[1-F(u)]^2 \ge u(1-2u).$$

Consequently this control gives a new state satisfying

$$q + 4q^2 \le q' \le q + 16q^2 \qquad (q \le p_0).$$

The scalar exponential potential from Lemma 3.1, at $t = 1$ and $a = 1$, gives entropy cost at least $2(q_j - q_{j-1})$ at each step. Summing and including the readout shows $V(q) \ge 1 - 2q$, so $V_* \ge 1$. In fact the inequality is strict in the limit. Let

$$g(q) = 2q - \log \mathbb{E} \exp\{2\tanh^2(\sqrt{q}Z)\}.$$

The strict heat-equation inequality gives $g(q) > 0$ for $q > 0$. Consider a path from $q < 1/100$ with cost at most two. If its final state is below $1/10$, its readout alone costs more than $5/4$. Otherwise consider its first crossing of $1/10$. The preceding state must exceed $1/100$: a transition with $q'/q \ge 10$ has entropy at least $[10 - 1 - \log 10]/2 > 2$. To see this, its preactivation second moment is at least q', since $\tanh^2 z \le z^2$; Gaussian entropy duality gives the lower bound $[r - 1 - \log r]/2$ at a second-moment ratio $r \ge 1$. The pre-crossing state therefore belongs to $[1/100, 1/10]$. Summing the strengthened step bounds $D \ge 2(q' - q) + g(q)$, and minimizing the terminal $2q_m + 1/(8q_m)$, gives total cost at least $1 - 2q + \min_{[1/100,1/10]} g$. The alternative final-state-below-$1/10$ case satisfies the same bound, since $\min_{[1/100,1/10]} g \le g(1/100) \le 1/50 < 1/4$. Paths of cost greater than two satisfy a stronger bound automatically. Letting $q \downarrow 0$ proves $V_* > 1$.

The same Gaussian-growth construction, stopped exactly at any $p \le p_0$, costs at most $17p$ when started at an arbitrarily small positive state. Indeed all full steps cost at most 16 times their variance increment. For the last step, continuously reduce the variance inflation to hit p exactly; its cost is at most $64p^2 \le p$. Appending a nearly optimal path from p and letting the starting state tend to zero yields $V_* \le 17p + V(p)$.

We also need a strict dissipation estimate. There is a universal $c > 0$ such that

$$\log \mathbb{E} \exp\{\tanh^2(\sqrt{q}Z)\} \le q - cq^2, \qquad 0 \le q \le 1. \tag{1}$$

For a direct verification, $G(z) = \exp(\tanh^2 z)$ satisfies $G'' < 2G$ away from zero. The heat equation gives a strict inequality $\log \mathbb{E}G(\sqrt{q}Z) < q$ for $q > 0$. Taylor expansion gives $\log \mathbb{E}G(\sqrt{q}Z) = q - q^2 + O(q^3)$ at zero. Compactness on $[0,1]$ therefore gives a positive c in (1). Entropy duality now implies

$$D(\eta_j \| \mathcal{N}(0, q_{j-1})) \ge q_j - q_{j-1} + cq_{j-1}^2. \tag{2}$$

An arbitrary finite path can be made strictly increasing without raising its cost. If a step decreases or preserves the state, delete that step. Start the remaining controls at the larger preceding state and rescale each successive control as in the monotonicity argument. Their entropies are unchanged and the final readout cost cannot increase. Repeat until there are no nonincreasing steps; the procedure terminates because each deletion shortens the path. Take a path within p of $V(p)$ and apply this procedure. The resulting increasing path has the asserted cost, proving the remaining claim. □

7.2 A small-variance layer forces the scalar cost

Lemma 7.3 (Comparison of a signed pair with the scalar control problem)**.** *For any admissible critical variational path and any signed input pair, put* $b = (e_x + se_y)/2$, $s \in \{-1, 1\}$, *and* $v_j = b^T Q_j b$. *If the output satisfies* $|b^T z| \geq 1/2$, *the part of the path after layer* j, *including the readout, costs at least* $V(v_j)$. *Moreover, a transition from pair variance* $v \leq 1$ *to* v' *has entropy cost at least* $v' - v + cv^2$, *with the constant from* (1).

Proof. It suffices to compare one transition. Lift a finite-entropy feature law through the inverse coordinatewise tanh map to its preactivation law. Under the reference law, the preactivation combination $b^T G$ is $\mathcal{N}(0, v)$. Under the controlled law, denote its distribution by η. Data processing bounds its Gaussian relative entropy by the original layer entropy. The pair contraction inequality gives

$$\int \tanh^2 z \, d\eta(z) \geq v',$$

where v' is the next feature-pair second moment. If the left side is q', the definition of V and its monotonicity imply

$$V(v) \leq D(\eta \| \mathcal{N}(0, v)) + V(q') \leq D(\nu' \| \mathsf{M}_Q) + V(v').$$

The final readout costs at least $1/(8v_{L-1}) \geq V(v_{L-1})$. Telescoping proves the assertion. Zero-variance or singular-covariance cases cause no problem: a finite-cost path reaching the stipulated nonzero pair output must have positive pair variance at every relevant step, and Cauchy–Schwarz is taken on the covariance range. For the final assertion, entropy duality applied to (1) gives

$$D(\eta \| \mathcal{N}(0, v)) \geq \int \tanh^2 z \, d\eta(z) - v + cv^2 \geq v' - v + cv^2.$$

Data processing transfers this bound to the full layer entropy. □

7.3 Holding and growing an arbitrarily weak target component

Suppose a feature second moment has the decomposition

$$Q = vff^T + R, \qquad R \succeq 0, \qquad R(x, x) = r \quad \text{for every } x.$$

Its diagonal is $q = v + r$. Inflate only the target component, taking the next preactivation covariance to be $Q + \epsilon vff^T$. Its entropy cost is at most $\epsilon^2/4$. Indeed, f lies in the range of Q, and $u = vf^T Q^+ f \leq 1$. The Gaussian relative entropy on that range is $[\epsilon u - \log(1 + \epsilon u)]/2 \leq \epsilon^2/4$; applying tanh cannot increase it. Put

$$\psi_r(z) = \mathbb{E} \tanh(z + \sqrt{r} Z), \qquad v' = \mathbb{E}\psi_r(\sqrt{(1+\epsilon)v} Z)^2, \qquad q' = F(q + \epsilon v).$$

Conditional expectation gives $Q' \succeq v' ff^T$. Conditional Gaussian Poincare, using the unit Lipschitz constant of tanh, also gives

$$r' := q' - v' \leq r. \tag{3}$$

The map $\epsilon \mapsto v'$ is continuous and nondecreasing. At $\epsilon = 0$ it is at most v, while the first Hermite projection gives

$$v' \geq (1 + \epsilon)v[1 - F(q + \epsilon v)]^2 \geq (1 + \epsilon)v[1 - 2(q + \epsilon v)]. \tag{4}$$

These statements depend only on the tracked scalars v, r, ϵ, not on the remaining entries of R or on the choice of f.

Lemma 7.4 (Cheap holding and prescribed growth)**.** *There are universal constants A, B, C_2 with the following properties. Let $D = d + 1$ and $0 < \delta \le p \le 1/128$. Starting from the unperturbed first feature law, at zero entropy cost and after at most B/δ hidden layers, one can track*

$$Q \succeq v_0 f f^T, \qquad v_0 = \delta e^{-AD}, \qquad Q(x,x) \le \delta.$$

One may hold this v_0 exactly fixed for any specified number H of layers, at total entropy cost at most $64H\delta^2$, retaining diagonal residual at most δ. Thereafter one can grow the tracked component exactly to p in at most BD/δ layers, with entropy cost at most

$$C_2 p + C_2 D\delta,$$

and with residual diagonal still at most δ. The number of growth layers is fixed in advance from v_0, δ, p, so it is independent of the preceding holding time and of f.

Proof. The first unperturbed covariance has smallest eigenvalue at least e^{-13D} by Lemma 4.1. Since the cube has $2^d \le e^D$ points, it dominates $e^{-14D} f f^T$ for every f. Lemma 6.1 shows that, after $O(1/\delta)$ unmodified layers, the diagonal lies in $[\delta/2, \delta]$ and the tracked target component is at least $b\delta e^{-14D}$. Enlarging A gives the displayed common lower component v_0; reduce the tracked component to precisely that value if necessary.

For holding, use an inflation $\epsilon \in [0, 16\delta]$ chosen to make $v' = v_0$. Such a choice exists by continuity: at zero inflation $v' \le v_0$, and (4) gives $v' \ge v_0$ at 16δ when $q \le \delta \le 1/128$. The residual does not increase, so the same argument applies at every holding layer. Each costs at most $64\delta^2$.

For growth, prescribe the scalar sequence

$$v_{j+1} = \min\{p,\ v_j[1 + 2(\delta + v_j)]\},$$

stopping upon reaching p. Throughout it, $r \le \delta$ and $v_j \le p$. At inflation $\epsilon = 16(\delta + v_j)$, (4) gives $v' \ge v_j[1 + 4(\delta + v_j)]$. Therefore an inflation no larger than this realizes the prescribed next value exactly. Its entropy cost is at most $64(\delta + v_j)^2$. For a full step, with $\Delta v = 2(\delta + v_j)v_j$, this cost is

$$32\Delta v + 32\delta \frac{\Delta v}{v_j}.$$

Here $\Delta v / v_j \le 4p < 1$, so $\Delta v / v_j \le 2\log(v_{j+1}/v_j)$. Summing, bounding the possible last step separately by $256p^2$, and using $\delta \log(p/\delta) \le p/e$, proves a bound $C_2 p + C_2 D\delta$.

Before v reaches δ, each full step multiplies it by at least $1 + 2\delta$, so at most $AD/\delta + 1$ steps are needed. Afterwards each full step decreases $1/v$ by at least one, so at most $1/\delta + 1$ further full steps and one final step are needed. This proves the stated BD/δ length bound. The prescribed scalar sequence fixes the length independently of the residual or holding time. □

7.4 Stable lifting of scalar controls

Lemma 7.5 (Lifting with a small residual)**.** *Suppose $Q \succeq p f f^T$ has constant diagonal $p + r$, with $0 \le r \le \delta$. Take a scalar path of length m starting at p. It can be lifted to a vector feature path with no greater total hidden entropy. Its final second moment dominates*

$$(1 - \delta)^{2m} q_m f f^T.$$

The residual diagonal stays at most δ throughout. If the scalar path has total cost at most three and $2m\delta \le 1/2$, the lifted path, including output $z = f/2$, costs at most the scalar cost plus $12m\delta$.

Proof. For $z \ge 0$ and $r \ge 0$, symmetry of a standard Gaussian gives

$$\psi_r(z) = \mathbb{E}\tanh(z + \sqrt{r}Z) \ge [1 - F(r)]\tanh z \ge (1 - r)\tanh z.$$

Indeed, writing $u = \tanh z$ and $w = \tanh(\sqrt{r}Z)$, the average of the two values corresponding to w and $-w$ is $u(1 - w^2)/(1 - u^2w^2) \ge u(1 - w^2)$. Oddness gives the corresponding absolute-value inequality for all real z.

Suppose the current tracked vector component is v, whereas the nominal scalar state is q. Rescale the nominal scalar preactivation control η by $\sqrt{v/q}$. Its entropy relative to $\mathcal{N}(0, v)$ equals the nominal entropy relative to $\mathcal{N}(0, q)$. Independently add a centered Gaussian with the residual covariance. Data processing bounds the vector feature entropy by that same scalar entropy. Its new tracked component is the second moment of the conditional feature mean. Since $\tanh(cz) \ge c\tanh z$ for $z \ge 0$ and $0 \le c \le 1$, the new component v' obeys

$$\frac{v'}{q'} \ge (1 - \delta)^2 \min\{1, v/q\},$$

where q' is the nominal next scalar state. The conditional-variance argument from (3) applies equally to a non-Gaussian scalar control, so the residual diagonal never increases. Starting from $v = q = p$ proves the component bound by induction.

The lifted readout cost is at most the nominal readout cost divided by $(1 - \delta)^{2m} \ge 1 - 2m\delta$. If $2m\delta \le 1/2$, its increase is at most $4m\delta$ times the nominal readout cost, which is at most three. This proves the final assertion. □

8 Short near-optimal scalar paths and residual spectral control

Near-optimality controls the entropy spent at every prefix of an increasing scalar path. This yields the path-length estimate used in the upper bound, and controls the spectral loss when the path is lifted to vector features. We use V, V_* and the constant $c > 0$ from (2).

Lemma 8.1 (Dyadic bounds for a near-optimal increasing path)**.** *There is a universal C such that, for every $0 < p \le 1/256$, an increasing scalar path from p with total cost at most $V(p) + p$ has length m and states $q_0 = p, q_1, \dots, q_m$ satisfying*

$$m \le C/p, \qquad \sum_{j=0}^{m-1} q_j \le C\log(2/p).$$

Such a path exists by the increasing-path reduction. The same sum bound holds with the post-step states $q_1, \dots, q_m$.

Proof. Write D_j for the entropy cost of step j. Every suffix from q_i including its readout costs at least $V(q_i)$, so

$$\sum_{j=1}^{i} D_j \le V(p) + p - V(q_i) \le V_* - V(q_i) + p. \tag{5}$$

Consider a dyadic band $[u, 2u)$, where $u = 2^k p < 1/256$. Because states increase, its transitions form one consecutive segment. At most one transition exits the band. Every other transition has post-state below $2u$, and the prefix ending at the last such transition has cost, by (5), at most

$$V_* - V(2u) + p \le 34u + p \le 35u.$$

Every step whose pre-state lies in this band has cost at least cu^2, by increasing states and strict dissipation. Thus at most $35/(cu)+1$ steps start in the band, contributing at most $70/c+2u$ to the sum of their pre-states.

Choose the last band so that its next lower endpoint lies in $[1/256, 1/128)$. All remaining pre-states are at least $1/256$. Since total cost is less than three, at most $3\cdot 256^2/c$ steps remain. Summing the dyadic counts gives $m \le C/p$, and summing the pre-state contributions gives $C\log(2/p)$. Finally, the post-state sum differs from the pre-state sum by $q_m - p \le 1$. □

Lemma 8.2 (A lower spectral bound for a lifted residual)**.** *Let $f \in \{-1,1\}^M$, let $R \succeq 0$ have constant diagonal r, and let $G \sim \mathcal{N}(0,R)$ be independent of a scalar random variable ξ. Define vector features*

$$h_x = \tanh(f_x\xi + G_x), \qquad \psi_r(z) = \mathbb{E}\tanh(z+\sqrt{r}Z),$$
$$v' = \mathbb{E}\psi_r(\xi)^2, \qquad q' = \mathbb{E}\tanh^2(\xi+\sqrt{r}Z), \qquad R' = \mathbb{E}hh^\top - v'ff^\top.$$

Then

$$R' \succeq (1-q')^2 R. \tag{6}$$

This applies to arbitrary scalar control laws with finite second moment, not only Gaussian controls.

Proof. Condition on ξ. The conditional mean of h is $f\psi_r(\xi)$. Put $a(\xi) = \mathbb{E}\operatorname{sech}^2(\xi+\sqrt{r}Z)$. Gaussian integration by parts gives

$$\operatorname{Cov}(h, G \mid \xi) = a(\xi)R.$$

The coefficient is the same at every input because the derivative is even and the residual diagonal is constant. The covariance matrix of $h - \mathbb{E}[h \mid \xi] - a(\xi)G$ is positive semidefinite, hence

$$\operatorname{Cov}(h \mid \xi) \succeq a(\xi)^2 R.$$

This argument also handles singular R by Gaussian approximation, or directly by a linear representation of G. Average over ξ and apply Jensen: $R' = \mathbb{E}\operatorname{Cov}(h \mid \xi) \succeq (\mathbb{E}a(\xi))^2 R = (1-q')^2 R$. □

Corollary 8.3 (Only polynomial spectral loss through a near-optimal lift)**.** *Lift a scalar path from Lemma 8.1 by the earlier scalar-lifting construction. Suppose the initial residual has constant diagonal at most $\delta \le p$. Then for universal $c_0 > 0, C_0 < \infty$, its final residual satisfies*

$$R_m \succeq c_0 p^{C_0} R_0.$$

Proof. At each step the scalar control is rescaled by the square root of the ratio of the tracked component v to the nominal state q. The tracked component never exceeds the nominal state. To check this, for $z \ge 0$ and $r \ge 0$, averaging the tanh addition formula at opposite Gaussian shifts gives

$$0 \le \psi_r(z) \le \tanh z.$$

If $v \le q$, rescaling by $\sqrt{v/q} \le 1$ can therefore only decrease the absolute scalar feature. Induction from $v = q = p$ proves the claim. The residual diagonal stays at most δ by conditional Gaussian Poincare. Thus the actual total feature variance at step j is at most $q_j + \delta$.

Every nominal step has entropy at most three and reference variance at most one. Gaussian entropy duality gives its preactivation second moment at most $12 + 2\log 2 < 14$. Rescaling does not increase this moment; adding the independent residual raises it by at most one. Consequently for each actual total post-step variance $\widehat{q}_j$,

$$1 - \widehat{q}_j = \mathbb{E}\operatorname{sech}^2(\xi+\sqrt{r}Z) \ge \mathbb{E}e^{-2|\xi+\sqrt{r}Z|} \ge e^{-2\sqrt{15}} =: a_0 > 0.$$

For $0 \le u \le 1 - a_0$, one has $\log(1-u) \ge -u/a_0$. Lemma 8.2 therefore gives

$$R_m \succeq \exp\left[-\frac{2}{a_0}\sum_{j=1}^{m}\widehat{q}_j\right] R_0 \succeq \exp\left[-\frac{2}{a_0}\left(\sum_{j=1}^{m} q_j + m\delta\right)\right] R_0.$$

The dyadic lemma bounds the state sum by $C\log(2/p)$ and $m\delta \le C$, proving the assertion. □

These are variational-path and population-covariance statements. They do not alone give a finite-width lower probability or a sample lower bound. Their role is to avoid an unnecessarily exponential spectral loss when constructing such a finite-width comparison.

9 The inverse-depth lower bound

The lower bound follows from the fact that a low-total-cost path has a small entropy budget at *every* prefix ending at a small variance.

Lemma 9.1 (A last-exit occupation bound)**.** *Let $q_0, \dots, q_H \in (0,1]$, with transition costs $D_j \ge 0$. Suppose there are universal constants $c, C_0 > 0$ such that*

$$D_j \ge q_j - q_{j-1} + cq_{j-1}^2, \qquad \sum_{j=1}^{i} D_j \le C_0 q_i \quad (0 \le i \le H), \qquad \sum_{j=1}^{H} D_j \le 2.$$

Then, for a constant C depending only on c, C_0,

$$H \le C/\epsilon, \qquad \epsilon = \min_{0\le j\le H} q_j.$$

The states need not be monotone.

Proof. Concavity of $q \mapsto -1/q$ gives

$$\frac{q_j - q_{j-1}}{q_{j-1}^2} \ge \frac{1}{q_{j-1}} - \frac{1}{q_j}.$$

On an interval of ℓ transitions whose pre-states are all at least $u > 0$, division of the dissipation inequality and summation therefore imply

$$c\ell \le \frac{\text{entropy spent on the interval}}{u^2} + \frac{1}{q_{\text{end}}} - \frac{1}{q_{\text{start}}}. \tag{7}$$

Fix $u_* = 1/256$. If $\epsilon \ge u_*$, apply this inequality to the entire path with total entropy at most two; its length is bounded by a universal constant, which is enough. Otherwise choose the least J such that $u_J = 2^J\epsilon \ge u_*$. Then $u_J < 2u_*$. For $0 \le j \le J$, let

$$\tau_j = \max\{i : q_i \le u_j\}.$$

These indices exist and are nondecreasing. The prefix ending at τ_0 ends at the global minimum ϵ, stays above it, and has entropy at most $C_0\epsilon$. By (7), its length is at most $(C_0+1)/(c\epsilon)$.

Consider a nonempty block from τ_j to τ_{j+1}. Remove its first transition temporarily. Every pre-state in the remaining interval exceeds u_j, by definition of τ_j, and its endpoint lies in $(u_j, 2u_j]$. The entropy of this interval is at most the total prefix entropy at τ_{j+1}, hence at most $2C_0u_j$. Its

length is consequently at most $(2C_0+1)/(cu_j)$. Restoring the first transition adds one. If the two last-exit indices coincide, the block is empty.

After τ_J, remove one first transition if present. Every remaining state exceeds $u_J \geq u_*$, and the total remaining entropy is at most two. The same weighted inequality bounds this final tail by a universal constant. Summing the geometric series $\sum_{j<J} u_j^{-1} \leq 2/\epsilon$ and absorbing the at most J removed transitions proves the assertion. □

Theorem 9.2 (Uniform inverse-depth lower bound)**.** *There is a universal constant C such that, for every input dimension $d \geq 1$, depth $L \geq 2$, and Boolean function f,*

$$J_L(f) \geq V_* - \frac{C}{L-1}.$$

Proof. For a nonconstant target, choose opposite labels on one cube edge; for a constant target, choose equal labels on an antipodal pair. In each case take the signed pair whose required output magnitude is at least $1/2$. Write $q_0 = 1/d$ and let $q_1, \dots, q_H$, $H = L-1$, be its successive pair variances. Every finite-cost path satisfying the required nonzero readout has positive pair variances. All lie in $(0,1]$. By Lemma 7.3, the full layer entropy costs satisfy the strict pair dissipation inequality required in Lemma 9.1.

Let A be the total path cost including the readout. If $A > V_*$, the desired conclusion is immediate. Otherwise the suffix comparison with the scalar control value gives, at every layer i,

$$\sum_{j=1}^{i} D_j \leq A - V(q_i) \leq V_* - V(q_i) \leq C_0 q_i.$$

The last inequality follows from the small-state bridge for $q_i \leq 1/128$ and from $0 \leq V(q_i) \leq V_* < 2$ for larger states; for example one may take $C_0 = 256$. Also $\sum_j D_j \leq A \leq V_* < 2$. Lemma 9.1 shows that the minimum pair variance ϵ is at most C/H. At a layer attaining it, the scalar suffix comparison gives

$$A \geq V(\epsilon) \geq V_* - C_0\epsilon \geq V_* - \frac{C'}{H}.$$

Take the infimum over admissible variational paths and rename the universal constant. □

10 The upper bound by rank-one purification

Lemma 10.1 (Purifying an exact target component)**.** *There are universal $p_0, c, C > 0$ with the following property. Suppose $Q = vff^T + R$, where $R \succeq 0$ has constant diagonal $0 \leq r \leq v \leq p_0$. One Gaussian covariance-inflation step of entropy at most Cv^2 produces a feature covariance*

$$Q' = vff^T + R', \qquad R' \succeq 0, \qquad R'(x,x) = r' \leq (1-cv)r.$$

Consequently T such steps have cost at most CTv^2, preserve the exact coefficient v, and reduce the residual diagonal by e^{-cvT}.

Proof. Inflate the target component by ϵvff^T. Write $\psi_r(a) = \mathbb{E}\tanh(a + \sqrt{r}Z)$ and

$$v'(\epsilon) = \mathbb{E}\psi_r(\sqrt{(1+\epsilon)v}Z)^2.$$

This is continuous, $v'(0) \leq v$, and the first-Hermite lower bound from the preceding construction gives

$$v'(\epsilon) \geq (1+\epsilon)v[1 - 2(v + r + \epsilon v)].$$

For sufficiently small $v + r$, its value at $\epsilon = 16(v + r)$ is at least v. Choose an intermediate ϵ with $v'(\epsilon) = v$. Its entropy cost is at most $\epsilon^2/4 \le 256v^2$.

The residual diagonal is the average conditional variance

$$r' = \mathbb{E}_A \operatorname{Var}_Z[\tanh(A + \sqrt{r}Z)], \qquad A \sim \mathcal{N}(0, (1 + \epsilon)v).$$

Gaussian Poincare gives

$$r' \le r\mathbb{E}\operatorname{sech}^4(\sqrt{u}Z), \qquad u = (1 + \epsilon)v + r.$$

The expansion $\mathbb{E}\operatorname{sech}^4(\sqrt{u}Z) = 1-2u+O(u^2)$ implies that it is at most $1-u$ for all sufficiently small u. Since $u \ge v$ and $u \le 2(v + r)$, reducing p_0 gives $r' \le (1 - v)r$. The conditional-mean covariance supplies the exact rank-one component vff^T; the remaining covariance is positive semidefinite and has the displayed constant diagonal. Iterate. □

11 The initial target component and the sufficient depth

The upper-bound construction can retain its dependence on the initial target component, rather than using the worst eigenvalue for every function. Let $Q_1 = Q(\mathsf{M}_{Q_0})$ be the first unperturbed feature covariance and define

$$a_f = (f^T Q_1^{-1} f)^{-1}, \qquad S_f = 1 + \log(1/a_f).$$

The matrix Q_1 is positive definite by Lemma 4.1. The rank-one inequality $Q_1 \succeq a_f ff^T$ holds by Cauchy–Schwarz, and its diagonal implies $0 < a_f \le 1$. Thus $S_f \ge 1$.

Theorem 11.1 (Target-dependent sufficient depth)**.** *There is a universal C such that, for every Boolean function f and integer depth $L \ge CS_f$,*

$$J_L(f) \le V_* + C\frac{S_f^2 + [\log(2L)]^2}{L}.$$

In particular, the uniform upper bound can be sharpened to

$$J_L(f) \le V_* + C\frac{(d+1)^2 + [\log(2L)]^2}{L} \qquad (L \ge C(d+1)).$$

Proof. The proof of Lemma 7.4 uses the dimension only through the logarithm of the initial tracked component. Starting from $Q_1 \succeq a_f ff^T$, the free-cooling lemma reaches diagonal in $[\delta/2, \delta]$ with tracked component at least $c\delta a_f$. Choose the smaller exact component $v_0 = \delta e^{-AS_f}$, with a fixed sufficiently large A. The same holding construction then has cost at most $64H\delta^2$ for any specified H. The prescribed growth sequence in that lemma reaches any $p \ge \delta$ in at most CS_f/δ layers, with cost at most $C(p + S_f\delta)$. Its length depends only on v_0, δ, p, and hence is independent of the holding interval. These assertions follow directly from the displayed growth sequence: below δ its length is $O(\delta^{-1}\log(\delta/v_0)) = O(S_f/\delta)$, and its entropy sum is $O(p + \delta\log(\delta/v_0))$.

Choose large universal constants B_1, B_2 and put

$$\delta = B_1 S_f/L, \qquad p = B_2\{S_f + \log(2L)\}/L.$$

Choose $B_2 \ge B_1$. The weaker depth requirement $L \ge CS_f$ already ensures $0 < \delta \le p \le p_0$ after choosing the universal C large enough: since $S_f \ge 1$ and $L \ge C$,

$$\delta \le B_1/C, \qquad p \le B_2\left\{C^{-1} + \frac{\log(2L)}{L}\right\} \le B_2\frac{1 + \log(2C)}{C}.$$

The last inequality uses monotonicity of $\log(2L)/L$ for large L. Here p_0 is the universal constant in Lemma 10.1. The total holding cost is allowed to be large; it remains explicitly in the final upper bound. Cooling and prescribed growth take less than $L/3$ layers if B_1 is large enough. Purify for $T = \lceil 4\log(2L)/(cp) \rceil$ layers; this leaves the exact component p and residual diagonal at most $\delta(2L)^{-4}$, at cost $O(p\log(2L))$. Append a scalar path of at most C_0/p steps and cost at most $V(p)+p$. Its stable lift adds only $O(\delta(2L)^{-4}/p) = O(L^{-4})$ readout cost. Purification and the scalar path together occupy less than $L/3$ layers if B_2 is sufficiently large. Fill the unused hidden layers by exact holding before prescribed growth. The number of inserted layers is an integer and all subsequent stage lengths are independent of it, so the final depth is exactly L.

The total excess over V_* is at most a universal multiple of

$$L\delta^2 + S_f\delta + p + p\log(2L) + L^{-4} \le C\frac{S_f^2 + [\log(2L)]^2}{L},$$

where $S_f \log(2L) \le \{S_f^2 + [\log(2L)]^2\}/2$. Finally, the first-layer spectrum bound and $\|f\|^2 = 2^d$ give $a_f \ge e^{-13(d+1)}/2^d \ge e^{-14(d+1)}$. Thus $S_f \le 15(d+1)$, yielding the uniform corollary. □

Corollary 11.2 (The depth scale for constants and dictators)**.** *There are universal constants $K_0, C > 0$ such that constants and dictators satisfy $J_L(f) \le K_0$ at every $L \ge C[\log(d+1)]^2$. More precisely,*

$$J_L(f) \le V_* + C\frac{[\log(d+1)]^2 + [\log(2L)]^2}{L} \qquad (L \ge C[\log(d+1)]^2).$$

Together with Theorem 20.1, this makes $\Theta((\log d)^2)$ the necessary and sufficient depth scale for attaining some fixed universal rate budget on these simplest targets. This is an absolute-rate statement, not a sample-complexity statement.

Proof. Write $q = (d+1)/d \in [1,2]$ and let $h_1(q)$ be the first normalized Hermite coefficient of $z \mapsto \tanh(\sqrt{q}z)$. Gaussian integration by parts gives

$$h_1(q) = \sqrt{q}\,\mathbb{E}\,\mathrm{sech}^2(\sqrt{q}Z) \ge c > 0 \qquad (1 \le q \le 2).$$

In the nonnegative Hermite expansion of Q_1, the first term is

$$\frac{h_1(q)^2}{d+1}\left(\mathbf{1}\mathbf{1}^T + \sum_{i=1}^{d} x_i x_i^T\right).$$

It follows that $Q_1 \succeq c(d+1)^{-1}ff^T$ for every constant or dictator, so $S_f \le C\log(d+1)$. Apply the theorem. The displayed excess is bounded by a universal constant on the specified depth range, after enlarging C, which proves the rate-budget assertion. The necessary depth follows from (14) with $k = 1$. □

Corollary 11.3 (The one-hidden-layer cost)**.** *There are universal $0 < c < C < \infty$ such that every Boolean target satisfies $J_2(f) \ge c\sqrt{d}$, whereas constants and dictators satisfy $J_2(f) \le C\sqrt{d}$.*

Proof. At $L = 2$, Theorem 20.1 gives $d \le 8J(1+\sqrt{2J})^2 \le 16J + 32J^2$. This implies $J \ge \sqrt{d}/64$ for every integer $d \ge 1$. For the constant target, shift only the first-layer Gaussian bias by $a = d^{-1/4}$. Its entropy cost is $da^2/2 = \sqrt{d}/2$. At every input the feature mean is the same positive number

$$b = \mathbb{E}\tanh(a + \sqrt{q}Z), \qquad q = (d+1)/d.$$

Since $0 < a \le 1$ and $1 \le q \le 2$, differentiation in a on this compact parameter range gives $b \ge c_0 a$ for a universal $c_0 > 0$. The resulting second moment dominates $b^2\mathbf{1}\mathbf{1}^T$, so the readout $z = \mathbf{1}/2$ costs at most $1/(8b^2)$. The total is at most $C\sqrt{d}$. Output sign symmetry and the augmented-coordinate permutation symmetry transfer the same bound to all constants and dictators. □

Corollary 11.4 (Functions of a small set of inputs)**.** *Suppose f depends on at most k of the d original input coordinates, where $0 \le k \le d$, and put*

$$R_k = (k+1)\log\frac{e(d+1)}{k+1}.$$

Then $S_f \le 14R_k$. In particular,

$$J_L(f) \le V_* + C\frac{R_k^2 + [\log(2L)]^2}{L} \qquad (L \ge CR_k^2).$$

Thus all such functions attain a fixed universal absolute rate budget at depth $O(R_k^2)$.

Proof. Set $D = d+1$ and $m = 2^d$. Let λ_T denote the eigenvalue of Q_1/m on the Walsh character χ_T, rather than the eigenvalue of the raw matrix Q_1. The Hermite proof of Lemma 4.1 gives, with $n = |T|$ if $|T|$ is odd and $n = |T|+1$ otherwise,

$$\lambda_T \ge e^{-12n}\frac{n!}{D^n}.$$

For $|T| \le k$, one has $1 \le n \le k+1 \le D$. The function $u \mapsto u\log(eD/u)$ is increasing on $(0, D]$, so

$$\lambda_T \ge \exp\{-12(k+1) - (k+1)\log(eD/(k+1))\}.$$

Write $f = \sum_T \widehat{f}(T)\chi_T$. Its nonzero coefficients have $|T| \le k$ and Parseval gives $\sum_T \widehat{f}(T)^2 = 1$. Since $Q_1\chi_T = m\lambda_T\chi_T$ and $\|\chi_T\|^2 = m$, the exact normalization is

$$f^T Q_1^{-1} f = \sum_T \frac{\widehat{f}(T)^2}{\lambda_T}.$$

The displayed coefficient bound implies $S_f \le 1 + 12(k+1) + R_k \le 14R_k$. Apply Theorem 11.1. □

12 Controlled Fourier seeds and the depth needed by parity

A weak component already present in the unmodified first-layer kernel can be amplified at small entropy cost. There is another option: change the first feature law slightly to insert a stronger component. For parity, this changes the sufficient depth from a degree-dependent bound to the same order as for constants and dictators. More generally, the Fourier ℓ^1 norm controls a useful family of initial components.

Use the normalized Fourier expansion on the Boolean cube,

$$f = \sum_{S\subseteq[d]} \widehat{f}(S)\chi_S, \qquad \chi_S(x) = \prod_{i\in S} x_i, \qquad A(f) = \sum_S |\widehat{f}(S)|.$$

For Boolean f, Parseval implies $A(f) \ge 1$.

Theorem 12.1 (A controlled Fourier-seed bound)**.** *There are universal constants C_0, C_1 such that, if*

$$T_f = \log(d + L + 2) + \log A(f), \qquad L \ge C_0 T_f,$$

then

$$J_L(f) \le V_* + C_1\frac{T_f^2}{L}.$$

In particular, a depth at least $C[\log(d+2) + \log A(f)]^2$ puts f below a fixed universal absolute critical-rate budget.

Corollary 12.2 (Parity and juntas)**.** *Every parity, irrespective of its degree, attains a fixed universal critical-rate budget at depth $O(\log^2(d+2))$. The universal lower bound in Theorem 20.1 makes $\Theta(\log^2(d+2))$ the necessary and sufficient depth scale for attaining some fixed universal budget, as d grows. Every function depending on at most k coordinates attains such a budget at depth $O((k+\log(d+2))^2)$. These are statements about prior rates, not sample-complexity guarantees.*

12.1 A common polynomial-cost family of parity laws

Lemma 12.3 (Parity mean and constant diagonal)**.** *For every $S \subseteq [d]$, there is a first-feature law ν_S with*

$$D(\nu_S \| \mathsf{M}_{Q_0}) \le D_* := C(d+1)^3 \log^2(d+2), \qquad \mathbb{E}_{\nu_S} H = c_S \chi_S, \qquad c_S \ge \frac{1}{8(d+1)}.$$

Its second-moment diagonal is constant and belongs to $[F(1), 1]$, where $F(q) = \mathbb{E}\tanh^2(\sqrt{q}Z)$ as above. Increase the universal C if necessary so that $D_* \ge 1$.

Proof. First construct a law with a signed mean margin, then symmetrize it. Let $k = |S| \ge 1$ and $u(x) = \sum_{i \in S} x_i$. At its possible values $u_j = -k + 2j$, the parity equals $f_j = (-1)^{k-j}$. Thus

$$\chi_S(x) = c_0 + \sum_{j=1}^{k} c_j \operatorname{sign}(u(x) - (-k+2j-1)), \qquad |c_0| \le 1, \quad c_j = (f_j - f_{j-1})/2 \in \{-1, 1\}.$$

Put $M = 4 + \log(64(k+1))$. Shift a first-row Gaussian weight vector to have mean coefficient M on each coordinate in S, zero on the other inputs, and bias $-M(-k+2j-1)$. Its preactivation mean has the required sign and absolute value at least M at every cube input. Its variance remains $q_0 = (d+1)/d \le 2$. For a centered Gaussian G of variance at most two,

$$1 - \mathbb{E}\tanh(t+G) \le 2e^{-2t+4} \qquad (t \ge 0).$$

The expected feature v_j therefore approximates its ideal step to maximum error $1/[16(k+1)]$. A constant bias shift supplies a constant mean $c_{\mathrm{const}} \ge 1/2$.

Take $\lambda = 1/[4(k+1)]$. Mix the appropriately signed detector laws with probabilities $\lambda|c_j|$, and the appropriately signed constant law with probability $\lambda|c_0|/c_{\mathrm{const}}$. Their sum is at most $\lambda(k+2) \le 1$; use the unmodified law for the remaining probability. Its feature mean is zero. The resulting mean m_S obeys

$$\chi_S(x) m_S(x) \ge \lambda \left(1 - \frac{k}{16(k+1)}\right) \ge \frac{1}{8(k+1)} \qquad \text{for every } x.$$

Every detector shift has Gaussian relative entropy at most

$$\frac{d}{2} M^2 \{k + (-k+2j-1)^2\} \le C(d+1)(k+1)^2 \log^2(k+2).$$

Convexity and data processing give the common bound D_* for this feature law. The case $k = 0$ uses only a positive constant bias shift, with constant positive mean and entropy $O(d)$.

For $b \in \{-1, 1\}^d$, transform features by

$$(T_b H)(x) = \chi_S(b) H(b_1 x_1, \ldots, b_d x_d).$$

Input weight sign changes and a simultaneous sign change of all weights implement this transformation. It preserves the raw first Gaussian law. Average the transformed controlled laws over b. Their

entropies are unchanged individually, so convexity bounds the mixture's entropy by D_*. Its mean is exactly

$$\mathbb{E}_b \chi_S(b) m_S(bx) = c_S \chi_S(x), \qquad c_S = \mathbb{E}_x \chi_S(x) m_S(x) \geq \frac{1}{8(k+1)}.$$

Transitivity of cube translations makes its diagonal second moment constant. Each original component was a Gaussian mean shift with unchanged covariance. For centered Gaussian G, $|G+t|$ stochastically dominates $|G|$, so $\mathbb{E}\tanh^2(G+t) \geq \mathbb{E}\tanh^2(G)$. Every diagonal before and after symmetrization is consequently at least $F(q_0) \geq F(1)$ and at most one. Only the first feature law has been symmetrized; no convexity of entire covariance trajectories is asserted. □

12.2 From a Fourier mixture to a small initial entropy cost

Proof of Theorem 12.1. Negating a whole feature preserves the raw first law and negates its mean. Choose the corresponding sign of each law in Lemma 12.3, and set

$$Z_f = \sum_S \frac{|\widehat{f}(S)|}{c_S} \leq 8(d+1)A(f).$$

Mix these laws with probabilities $|\widehat{f}(S)|/(c_S Z_f)$. The resulting law ν_f has entropy at most D_*, mean exactly f/Z_f, and second moment

$$Q(\nu_f) \succeq Z_f^{-2} f f^T.$$

Its diagonal is constant in $[F(1), 1]$. The component diagonals may have different values, but each is constant over inputs. Convexity introduces no factor for the number of Fourier components.

Now replace the original first law by

$$\widetilde{\nu}_f = (1-\theta)\mathsf{M}_{Q_0} + \theta \nu_f, \qquad \theta = (D_* L)^{-1}.$$

Its entropy is at most $1/L$, its diagonal remains constant in $[F(1), 1]$, and

$$Q(\widetilde{\nu}_f) \succeq v f f^T, \qquad v = \frac{1}{D_* L Z_f^2}.$$

This uses the second moment of the mixture; it does not square the mixing probability. Hence

$$S_v := 1 + \log(1/v) \leq C[\log(d + L + 2) + \log A(f)] = C T_f.$$

We spell out the initial-condition adjustment in the existing steering argument. Lemma 6.1 is stated for an initial diagonal in $[F(1), F(2)]$, but its proof works on $[F(1), 1]$. Indeed $F(q) < q$, $(q - F(q))/q^2$ has a positive lower bound on $(0, 1]$, and

$$R(q) = \frac{q[1-F(q)]^2}{F(q)} > 0, \qquad \log R(q) = O(q^2) \quad (q \downarrow 0).$$

Compactness extends the required estimates to the enlarged interval. The same telescoping proof reaches diagonal δ in $O(1/\delta)$ free layers while retaining a component at least $c\delta v$.

The proof of Theorem 11.1 then applies with S_v in place of S_f and with the first-law entropy $1/L$ added. For clarity, set $\delta = B_1 S_v / L$ and $p = B_2(S_v + \log(2L))/L$ with the sufficiently large universal constants used there. Prescribed rank-one growth costs $O(p + S_v\delta)$ and uses $O(S_v/\delta)$ layers. Exact holding in the unused layers costs $O(L\delta^2)$. Purification costs $O(p \log(2L))$, leaves residual $O(\delta L^{-4})$,

and the scalar path costs at most $V(p)+p \le V_*+p$, with $O(L^{-4})$ additional readout cost. All stage lengths fit when $L \ge CS_v$, and holding gives the exact specified depth. Adding the initial entropy yields

$$J_L(f) \le V_* + \frac{1}{L} + C\frac{S_v^2 + \log^2(2L)}{L} \le V_* + C_1\frac{T_f^2}{L}.$$

The strict-margin rate equals its closed-margin variational infimum by Corollary A.2, so this construction has precisely the prior-rate interpretation in the theorem.

Finally let $B = \log(d+2) + \log A(f) \ge 1$. One has $T_f \le B + \log(L+1) + C$. For every $L \ge CB^2$, increasing C if necessary makes T_f/L satisfy the depth requirement and bounds T_f^2/L by a universal constant. This proves the fixed-budget assertion. □

Proof of Corollary 12.2. A parity has $A(f) = 1$. For a k-junta the Fourier support has at most 2^k elements, so Parseval and Cauchy–Schwarz give $A(f) \le 2^{k/2}$. Substitute these bounds into the theorem. The matching necessary depth for any fixed rate budget follows from Theorem 20.1. □

The controlled mixture explains why a small eigenvalue in the original first-layer kernel cannot by itself give a large critical-rate lower bound. The mixture inserts a stronger component at entropy $1/L$. The theorem gives no lower bound ordering the competing extensions of a training sample; the next section isolates that remaining question.

13 Exact Boolean feature means with a common diagonal

The next construction propagates exact Boolean gate vectors through a controlled feature law. Both the mean and the diagonal second moment are exact. In particular, approximation errors do not accumulate across the circuit layers. The number of mixture components can be exponential in the state width; only its logarithm enters the entropy bound.

Lemma 13.1 (Exact finite-state feature repair)**.** *Let X be a finite set, and let $H : X \to \{-1,1\}^s$ contain a constant coordinate $H_0 = 1$. Regard H also as the matrix whose rows are indexed by X. Suppose*

$$Q \succeq rHH^{\mathsf{T}}, \qquad r > 0, \qquad Q(x,x) = q \le 2 \quad (x \in X).$$

Let $g(x) = \operatorname{sign}(w \cdot H(x))$, where $|w \cdot H(x)| \ge 1$ everywhere. There is a feature law ν_g such that

$$\mathbb{E}_{\nu_g} U(x) = ag(x), \qquad \mathbb{E}_{\nu_g} U(x)^2 = q_* \quad (x \in X), \qquad a \ge \tfrac{1}{2}, \quad q_* \ge \tfrac{1}{2},$$

and

$$D(\nu_g \| \mathsf{M}_Q) \le \frac{C(s+1)^2\{\|w\|_2^2 + (s+1)^2\}}{r}.$$

Here $\mathsf{M}_Q = \tanh_* N(0,Q)$, and C is universal. For fixed H and Q, the same a and q_* work for every gate g. The law ν_g is a finite mixture of Gaussian mean shifts followed by coordinatewise tanh*. The covariance Q may be singular.*

Proof. For every coefficient vector c, the covariance inequality gives $Hc \in \operatorname{range} Q$ and

$$(Hc)^{\mathsf{T}} Q^{+} (Hc) \le \|c\|_2^2 / r.$$

Indeed, $rQ^{+/2}HH^{\mathsf{T}}Q^{+/2} \preceq I$ on the covariance range, which bounds the operator norm of $\sqrt{r}\, Q^{+/2}H$ by one. A Gaussian mean shift by Hc therefore has relative entropy at most $\|c\|_2^2/(2r)$. The same upper bound holds after applying tanh.

Write $T = H(X)$ for the distinct realized states and $M = |T|$. The constant coordinate implies $M \le 2^{s-1}$. Set

$$F(q) = \mathbb{E}\tanh^2(\sqrt{q}Z), \qquad c_* = 1 - F(2) > 0, \qquad \varepsilon = \frac{c_*^2}{10^6 M}.$$

Choose $A \ge 1$ with $4e^{-2A+4} \le \varepsilon$ and $A \le C(s+1)$. If a Gaussian variable has variance at most two and mean of absolute value at least A, its expected tanh differs from the mean's sign by at most $\varepsilon/2$, and its second moment belongs to $[1-\varepsilon, 1]$. For positive means this follows from

$$1 - \tanh u \le 2e^{-2u}, \qquad \operatorname{sech}^2 u \le 4e^{-2u};$$

negative means follow by symmetry.

For each $p \in T$ define

$$d_p(H) = \frac{s - p \cdot H}{2}.$$

This is zero at state p, is a positive integer at every other state, and is at most s. Using the constant coordinate, both $d_p(H)$ and $1 - 2d_p(H)$ are linear combinations of the coordinates of H with coefficient norm at most Cs.

We use the following feature laws, all obtained from a Gaussian of covariance Q by a mean shift and then tanh:

- P_g has preactivation mean $Aw \cdot H$. Write its feature mean as m_g; then $\|m_g - g\|_\infty \le \varepsilon/2$.
- P_p has preactivation mean $A(1 - 2d_p(H))$. Its feature mean approximates $2\mathbf{1}_{\{H=p\}} - 1$ to error at most $\varepsilon/2$.
- P_c has the constant preactivation mean As. Write its constant feature mean as m_c and second moment as s_c.
- R_c is the equal mixture of P_c and its output-negated law; its mean is zero and its second moment is s_c.
- R_p is the equal mixture of the shifts with preactivation means $Ad_p(H)$ and $-Ad_p(H)$. Its mean is zero. Its second moment is $F(q)$ at state p and belongs to $[1-\varepsilon, 1]$ elsewhere.

The diagonals of P_g, P_p, P_c, R_c all belong to $[1-\varepsilon, 1]$. Each law above has entropy at most

$$E = \frac{CA^2(\|w\|_2^2 + s^2)}{r}.$$

For the mixtures this follows from entropy convexity. Output negation preserves the raw law and is realized by negating the mean shift, so it preserves these entropy bounds. Because Q has constant diagonal, every mean and second moment just described depends only on the state $H(x)$.

Repairing the mean. Let $B_p = (P_p + P_c)/2$. Its mean on T approximates the point indicator of p to error at most $\varepsilon/2$. The square matrix V with these means as columns consequently satisfies

$$\|V - I\|_{\infty\to\infty} \le M\varepsilon/2 < 1/2, \qquad \|V^{-1}\|_{\infty\to\infty} \le 2.$$

Put $t_1 = 4M\varepsilon$ and solve

$$Vc = (1 - t_1)(g - m_g).$$

Then $\|c\|_\infty \le \varepsilon$ and $\sum_p |c_p| \le M\varepsilon < t_1$. Define

$$\nu_1 = (1-t_1)P_g + \sum_p |c_p| \operatorname{sign}(c_p) B_p + \Big(t_1 - \sum_p |c_p|\Big) R_c,$$

where multiplication of a law by -1 denotes its pushforward by output negation; terms with $c_p = 0$ are omitted. This is a probability mixture. Its mean is exactly $(1-t_1)g$, its diagonal $s_1(x)$ belongs to $[1-\varepsilon, 1]$, and its entropy is at most E.

Repairing the diagonal with positive weights. Let $u_p(x)$ be the second moment of R_p, and on the state set T form the matrix

$$D_{x,p} = s_c - u_p(x).$$

For a centered Gaussian, the absolute value after a mean shift is stochastically increasing in the absolute mean. Since $|Ad_p(H)| \le As$, the entries of D are nonnegative. Its diagonal is the common number

$$c = s_c - F(q) \ge c_* - \varepsilon \ge c_*/2,$$

and every off-diagonal entry is at most ε. Thus $D = cI + E_0$ with $\|E_0\|_{\infty\to\infty} \le M\varepsilon$.

Set

$$t_2 = 32M\varepsilon/c_*, \qquad b_x = 4\varepsilon - (1-t_2)(s_c - s_1(x)).$$

Both t_1, t_2 lie in $(0, 1/10)$, and $3\varepsilon \le b_x \le 5\varepsilon$. Let $\lambda = D^{-1}b$. With $\delta = \|E_0\|_{\infty\to\infty}/c \le 2c_*/10^6$, the Neumann series gives

$$\|\lambda - b/c\|_\infty \le \frac{\delta}{1-\delta}\frac{5\varepsilon}{c}.$$

The right side is less than $3\varepsilon/(10c)$, so every λ_p is strictly positive. Moreover,

$$\|\lambda\|_\infty \le \frac{5\varepsilon}{c(1-\delta)} \le \frac{12\varepsilon}{c_*}, \qquad \sum_p \lambda_p < t_2.$$

The probability mixture

$$\nu_g = (1-t_2)\nu_1 + \sum_p \lambda_p R_p + \Big(t_2 - \sum_p \lambda_p\Big) R_c$$

has exact mean ag, where $a = (1-t_1)(1-t_2) \ge 1/2$, and exact diagonal

$$(1-t_2)s_1 + t_2 s_c - D\lambda = s_c - 4\varepsilon =: q_* \ge 1/2.$$

Neither a nor q_* depends on the gate. Entropy convexity gives $D(\nu_g \| \mathsf{M}_Q) \le E$. Finally, $A \le C(s+1)$ gives the asserted polynomial numerator. □

14 Controlled feature laws for arbitrary Boolean circuits

The Fourier construction can be replaced by a layerwise simulation of a Boolean circuit. The controlled laws used in this simulation are rare mixture components. Their probabilities become small with circuit depth, but the logarithm of the resulting target component grows only linearly with that depth. An exact repair of feature means and diagonal second moments makes this compatible with the subsequent scalar steering. This section bounds the critical prior rate; it does not establish a sample-complexity guarantee.

Definition 14.1 (Layered Boolean threshold implementation)**.** *Let $H^{(0)}(x) = (1, x_1, \dots, x_d)$. A depth-$h$ implementation consists of state vectors $H^{(j)} : \{-1,1\}^d \to \{-1,1\}^{C_j}$, $1 \le j \le h$, each containing a constant coordinate, with*

$$H_i^{(j)}(x) = \operatorname{sign}(w_{j,i} \cdot H^{(j-1)}(x)), \qquad |w_{j,i} \cdot H^{(j-1)}(x)| \ge 1 \quad \text{for every } x.$$

One coordinate of $H^{(h)}$ is the target f. Write $C = \max_j C_j$, $W = \max\{d+1, C\}$, and assume $\|w_{j,i}\|_2 \le B$ with $B \ge 1$. The constant coordinate allows affine gates without adding internal biases to the neural-network model. A bounded-fan-in AND/OR/NOT circuit has such an implementation with universal B, after a fixed scaling of each gate's coefficients.

Theorem 14.2 (A circuit-dependent critical-rate bound)**.** *There are universal constants C_0, C_1, C_2 with the following property. Suppose f has the implementation above, with $h \ge 1$, and put*

$$T = \log(d+2) + h \log(W + B + h + L + 2).$$

If the learner depth satisfies $L - h \ge C_0 T$, then

$$J_L(f) \le V_* + C_1 \frac{T^2}{L-h}.$$

In particular, every

$$L \ge C_2 h^2 \log^2(W + B + h + 2)$$

puts f below a fixed universal absolute critical-rate budget. The prior remains the original independent Gaussian prior; no robustness restriction is imposed on its support.

Corollary 14.3 (Bounded-fan-in Boolean circuits)**.** *For a layered bounded-fan-in AND/OR/NOT circuit of width C and depth $h \ge 1$, including constant rails, define*

$$T_{\mathrm{B}} = \log(d+2) + h \log(C + h + L + 2).$$

If $L - h \ge C_0 T_{\mathrm{B}}$, then

$$J_L(f) \le V_* + C_1 \frac{T_{\mathrm{B}}^2}{L-h}.$$

Consequently every

$$L \ge C_2 [\log(d+2) + h \log(C + h + 2)]^2$$

puts f below a fixed universal absolute critical-rate budget. The constants may depend on the fixed fan-in bound and gate library.

The local ingredient is Lemma 13.1 below. It constructs an exact Boolean feature mean and an exactly constant diagonal at polynomial entropy cost. The exponentially many states used in its repair do not cause exponential cost: each component has polynomial entropy cost, and convexity bounds the cost of their mixture by the same bound. Individual repair weights are exponentially small; their total mass is a small constant.

14.1 A low-cost circuit prefix

Lemma 14.4 (Circuit prefix)**.** *For every $L \ge h + 1$, the first h hidden feature laws can be chosen at total relative entropy at most $1/L$, so that their final second moment has constant diagonal in $(0, 1]$ and satisfies*

$$Q_h \succeq r_h H^{(h)} (H^{(h)})^T, \qquad 1 + \log(1/r_h) \le C[\log(d+2) + h \log(W + B + h + L + 2)].$$

Proof. The input covariance has the exact representation

$$Q_0 = d^{-1}H^{(0)}(H^{(0)})^T, \qquad Q_0(x,x) = (d+1)/d \le 2.$$

Set $r_0 = 1/d$, and choose the universal polynomial

$$P = C_*(W+1)^2\{B^2 + (W+1)^2\} \ge 1$$

large enough to bound the numerator in Lemma 13.1 for every gate. Suppose Q_{j-1} has constant diagonal and dominates $r_{j-1}H^{(j-1)}(H^{(j-1)})^T$. For each gate in the next state, that lemma gives a feature law with mean $a_j H_i^{(j)}$, where $a_j \ge 1/2$, and entropy at most P/r_{j-1} relative to $\mathsf{M}_{Q_{j-1}}$. For this fixed state and reference covariance, all the laws have the same a_j and the same constant diagonal.

Average these controlled laws uniformly over the C_j gates. Their mixture ν_j has constant diagonal and satisfies

$$Q(\nu_j) \succeq \frac{1}{4C_j}H^{(j)}(H^{(j)})^T, \qquad D(\nu_j \| \mathsf{M}_{Q_{j-1}}) \le \frac{P}{r_{j-1}}.$$

Use the diluted law

$$\widetilde{\nu}_j = (1-\theta_j)\mathsf{M}_{Q_{j-1}} + \theta_j \nu_j, \qquad \theta_j = \frac{r_{j-1}}{PhL}, \qquad r_j = \frac{r_{j-1}}{4C_j PhL}.$$

Here $0 < \theta_j \le 1$ because $r_{j-1} \le 1$. Entropy convexity bounds the layer cost by $1/(hL)$, and its second moment dominates $r_j H^{(j)}(H^{(j)})^T$. The uncontrolled law has constant diagonal whenever its reference does, so dilution preserves this property. Every hidden diagonal is at most one, and its constant rail makes it strictly positive. Induction therefore applies through all h layers. Finally,

$$\log(1/r_h) = \log d + \sum_{j=1}^{h} \log(4C_j PhL) \le \log d + h\log(4WPhL),$$

which gives the stated estimate. □

14.2 Steering from an arbitrary positive constant diagonal

Lemma 14.5 (Steering after a prescribed prefix)**.** *Suppose a prefix of h hidden layers has entropy cost E and ends at a second moment Q with constant diagonal $q \in (0,1]$ and $Q \succeq vff^T$, where $0 < v \le 1$. Write $S = 1 + \log(1/v)$ and $R = L - h$. For $R \ge CS$, it can be completed to the exact learner depth L with*

$$J_L(f) \le E + V_* + C\frac{S^2 + \log^2(2R)}{R}.$$

Proof. Only the initial cooling step of Theorem 11.1 needs adjustment. For $q \in (0,1]$, the estimates in Lemma 6.1 hold uniformly: $q - F(q) \ge cq^2$, and the product of the factors

$$\frac{q[1-F(q)]^2}{F(q)}$$

along a free trajectory is bounded below by a universal positive constant. The latter follows from a logarithm of order $O(q^2)$ at zero, positivity on $(0,1]$, and the uniformly bounded sum of q_j^2. If the current diagonal exceeds a desired level δ, cool to the first diagonal below δ. This takes $O(1/\delta)$

layers, ends with diagonal at least $c\delta$, and retains a tracked component at least $c\delta v$; the initial diagonal is at most one. If $q \le \delta$ already, skip cooling. Its existing component v is also at least $c\delta v$.

In either case reduce the tracked component to the exact value $v_0 = \delta e^{-AS}$ for a sufficiently large universal A. The residual diagonal is at most δ. The holding and prescribed-growth arguments of Lemma 7.4 use precisely this upper bound; they do not require an initial diagonal bounded below by δ. Consequently holding costs $O(H\delta^2)$ for a prescribed H, and growth to $p \ge \delta$ takes $O(S/\delta)$ layers with entropy $O(p + S\delta)$.

Choose $\delta = B_1 S/R$ and $p = B_2\{S + \log(2R)\}/R$, as in the target-component theorem. Its purification, stable scalar lifting, and exact-depth padding then fit among the remaining $R-1$ hidden layers for $R \ge CS$, after increasing the universal constants. The cost after the prefix is at most

$$V_* + C\{R\delta^2 + S\delta + p + p\log(2R) + R^{-4}\} \le V_* + C\frac{S^2 + \log^2(2R)}{R}.$$

The prefix cost adds E, as claimed. □

Proof of Theorem 14.2. Apply Lemma 14.4 and retain the coordinate corresponding to f, giving $Q_h \succeq r_h f f^T$ with $S = 1 + \log(1/r_h) \le CT$. Apply Lemma 14.5 with $E \le 1/L$ and $R = L - h$. Since $h \ge 1$, the definition of T also absorbs $\log(2R)$. This proves the displayed rate bound. The equality between strict margin rates and their variational infima gives its prior-probability interpretation.

For the fixed-budget claim, put $K = W + B + h + 2$ and note $K \ge h + 2$. At $L \ge Ch^2 \log^2 K$, increasing C ensures $L - h \ge L/2$, the required depth condition, and a universal bound for $h^2[\log K + \log L]^2/L$. Indeed $(\log K + \log L)^2/L$ is decreasing on this range, and its value at the lower endpoint is $O(h^{-2})$. The rate bound is therefore uniformly bounded there. □

Proof of Corollary 14.3. For each gate apply Lemma 13.1 only to its input wires and the constant coordinate. This restricted state has bounded width, its threshold coefficient norm is universally bounded, and $Q \succeq rHH^T$ implies domination of the restricted state's second moment with the same r. The controlled gate law therefore costs at most P_*/r with universal P_*. Its mean amplitude is at least $1/2$ and its diagonal is constant. These constants need not agree between different gates: a mixture of laws whose diagonals are individually constant still has constant diagonal, and the uniform amplitude lower bound suffices for covariance domination.

Repeat the prefix proof with $P = P_*$ and $C_j \le C$. This gives

$$1 + \log(1/r_h) \le 1 + \log d + h\log(4CP_*hL) \le C_* T_{\mathrm{B}}.$$

The same steering proves the rate bound. For the final assertion put $A = \log(d+2) + h\log(C+h+2)$. Then $A \ge h\log(h+2)$ and $T_{\mathrm{B}} \le A + h\log(L+1) + C_* h$. At $L = C_2 A^2$, one has $h\log(L+1) = O(A)$, apart from the harmless constant $h\log C_2$, since $h\log A = O(A)$ for $A \ge h\log(h+2)$. Increasing C_2 gives the depth requirement and a universal bound on $T_{\mathrm{B}}^2/(L-h)$ at that point. The bound persists for all larger L: the function $(A + h\log(L+1) + C_* h)^2/L$ is decreasing once $A + h\log(L+1) + C_* h > 2h$, and $L - h \ge L/2$ on the specified range. □

Combining this construction with the Fourier and unmodified-kernel constructions permits the best of their upper bounds for any particular target. The new construction avoids exponential dependence on the teacher's depth in the critical prior-rate upper bound. A learning theorem still requires a comparison with inaccurate sample-consistent functions; the near-baseline posterior ceiling does not provide that comparison by itself.

15 Boolean functions implemented by real-valued tanh teachers

The circuit-rate bound also applies to Boolean targets specified by a bounded-weight tanh teacher. Finite precision is enough because the teacher's final output has a fixed margin. The necessary precision grows linearly with teacher depth, and each numerical layer has a constant-depth threshold implementation of polynomial size. No stability condition on the teacher's hidden states is needed.

Lemma 15.1 (Finite-precision threshold compilation)**.** *There is a universal integer a with the following property. Suppose a tanh network on $\{-1,1\}^d$ has depth $D \geq 1$, hidden width at most C, parameter magnitudes at most $B \geq 1$, and a linear readout z_T satisfying*

$$f(x)z_T(x) \geq \tfrac{1}{2} \qquad \text{for every } x, \qquad f : \{-1,1\}^d \to \{-1,1\}.$$

Depth counts affine maps, including the linear readout. Biases may be present; the original model with a constant input and no internal biases is a special case. Then f has a layered Boolean threshold implementation of depth $h \leq a(D+1)$, width at most K^a, and gate coefficient Euclidean norm at most K^a, where

$$K = d + C + B + D + 2.$$

Every Boolean gate has margin at least one, and constant rails and layering overhead are included in these bounds.

Proof. We use the standard arithmetic threshold-circuit theorem: addition, comparison, division, and iterated multiplication of polynomially many polynomial-bit integers have constant-depth, polynomial-size circuits of unweighted majority and Boolean gates; see Hesse–Allender–Barrington [1] for the nontrivial division and iterated-multiplication assertions. The depth and polynomial degree are universal. Thus this theorem provides majority gates, rather than arbitrary threshold gates with uncontrolled coefficient magnitudes.

First consider approximating a single scalar tanh. For a dyadic input u with $|u| \leq R$ and desired error 2^{-p}, put $t = 2|u|$ and

$$E_m(t) = \sum_{j=0}^{m-1} \frac{t^j}{j!}, \qquad m = \lceil 40(R+p+2) \rceil.$$

The Taylor remainder satisfies

$$0 \leq e^t - E_m(t) \leq e^{2R} \frac{(2R)^m}{m!} \leq e^{2R} \left(\frac{2eR}{m}\right)^m < 2^{-p-4}.$$

Write the sum over a common denominator. All its powers, factorial constants, integer products, and sums involve polynomially many bits in R, p, and the input bit length. They can therefore be computed in constant threshold depth by the arithmetic theorem. Since $E_m(t) \geq 1$, the map $v \mapsto (v-1)/(v+1)$ has derivative at most one for $v \geq 1$. Division followed by rounding to $p+2$ fractional bits gives

$$\operatorname{sign}(u) \frac{E_m(2|u|) - 1}{E_m(2|u|) + 1}$$

to total error less than 2^{-p} from $\tanh u$. Choose the rounding convention to keep the result in $[-1,1]$. This is a constant-depth majority circuit whose size is polynomial in the stated numerical parameters, with universal degree.

For the teacher, set

$$K_0 = 4(d + C + 1)(B + 1), \qquad p = \lceil (D + 3) \log_2 K_0 \rceil + 5.$$

Round every parameter to p fractional bits and hardcode these bits in the circuit. This is a nonuniform existence statement and requires no computability assumption on the original real parameters. Starting from the exact Boolean input, evaluate each affine map on the stored dyadic values exactly, and evaluate each tanh to error at most 2^{-p} by the preceding construction. All stored hidden values have magnitude at most one, so every preactivation has magnitude at most $R = (d + C + 1)(B + 1)$. Parameter rounding contributes at most $(d + C + 1)2^{-p}$ at any layer. Since tanh is one-Lipschitz, a conservative induction bounds the error after j affine maps by $K_0^{j+1} 2^{-p}$, including the linear readout. The final error is therefore less than $1/16$. Its sign is exactly f by the margin assumption.

The arithmetic at every layer has constant threshold depth and polynomial size in d, C, B, D: the precision p, the range R, all intermediate bit lengths, and the number of Taylor terms are polynomial in these parameters. Stack these blocks over D layers and append a sign comparison. The resulting depth is at most $a(D + 1)$ and its total size is at most K^a after increasing a universal a. Make it layered by identity gates and propagate a constant wire. This introduces only polynomial overhead and a universal constant factor in depth.

Finally, an unweighted majority or Boolean gate with fan-in at most the total circuit size can be written as an integer threshold gate with coefficients of polynomial magnitude and margin at least one. For example, place the threshold halfway between the two neighboring allowed integer sums, and multiply the score by two if necessary. The constant rail carries the threshold. Its coefficient norm is polynomial in the circuit size. Increasing a once more proves all three bounds simultaneously. □

Corollary 15.2 (Polynomial teacher-depth dependence of the critical rate)**.** *For a Boolean target with the tanh-teacher implementation in Lemma 15.1, there is a universal A such that, whenever*

$$L \geq A(D + 1) \log(d + C + B + D + L + 2),$$

the original unrestricted Gaussian learner satisfies

$$J_L(f) \leq V_* + A \frac{(D + 1)^2 \log^2(d + C + B + D + L + 2)}{L}.$$

In particular, every

$$L \geq A(D + 1)^2 \log^2(d + C + B + D + 2)$$

puts f below a fixed universal absolute critical-rate budget.

Proof. Apply the compiler lemma. Its threshold depth is $h \leq a(D + 1)$, while its state width and normalized gate coefficient norms are polynomial in $K = d + C + B + D + 2$. The parameter T in Theorem 14.2 is consequently at most $A_0(D + 1) \log(K + L)$. Increase the universal A so that the displayed depth condition ensures $L - h \geq L/2$ and $L - h \geq C_0 T$. The theorem gives the rate estimate. Its fixed-budget conclusion, or the same squared-logarithm comparison used in its proof, gives the final assertion. □

The result concerns the rate of the target's full Boolean sign pattern under the original Gaussian prior. It removes an exponential teacher-depth cost from this rate upper bound. It does not prove that posterior conditioning on polynomially many examples favors that target over inaccurate extensions, and therefore does not settle the general sample-complexity question.

16 A sample-dependent posterior ceiling near the common rate

The Fourier construction supplies a low-rate, globally separated extension of *every* consistently labeled sample. Thus it gives an actual-posterior comparison, rather than only a representation bound.

Theorem 16.1 (Near-baseline critical posterior ceiling)**.** *There are universal constants C_0, C_1 such that the following holds. Let S be any consistently labeled sample of at most $n \geq 1$ Boolean inputs, and suppose*

$$L \geq C_0 \log(2 + d + n + L).$$

Put

$$U_{d,n,L} = V_* + C_1 \frac{\log^2(2 + d + n + L)}{L}.$$

For the original globally margin-conditioned critical posterior,

$$\Pi_{N,1}\{g : J_L(g) > U_{d,n,L} \mid S\} \longrightarrow 0 \qquad (N \to \infty).$$

The dimension, depth, and realized sample are fixed in this limit. The constants and the rate ceiling are uniform over the sample.

Proof. Let $p_1, \ldots, p_m$ be the distinct negatively labeled sample points, where $m \leq n$. The Boolean function

$$g_S(x) = 1 - 2 \sum_{j=1}^{m} \mathbf{1}_{\{x = p_j\}}$$

agrees with the sample, including any repeated inputs. A point indicator has normalized Fourier ℓ^1 norm exactly one, since

$$\mathbf{1}_{\{x=p\}} = 2^{-d} \prod_{i=1}^{d} (1 + p_i x_i).$$

Therefore $A(g_S) \leq 1 + 2m \leq 1 + 2n$. Theorem 12.1 gives $J_L(g_S) \leq U_{d,n,L}$ after adjusting universal constants.

At fixed d, there are finitely many Boolean functions. Their exact prior rates and the posterior-rate formula are given by Corollary A.2. The denominator's rate is at most $J_L(g_S)$. If any consistent functions have rate greater than $U_{d,n,L}$, the finite set of their rates has a strictly positive gap above this ceiling. Their total posterior mass consequently tends to zero. If there are none, the assertion is immediate. This argument places no restriction on mixtures of tied minimizers. □

Combining this ceiling with the universal lower bound gives the limiting rate interval

$$V_* - C/L \leq J_L(g) \leq V_* + C \frac{\log^2(2 + d + n + L)}{L}$$

for posterior-typical functions, in the stated fixed-domain width limit. Small remaining differences inside this interval can still determine learning.

Corollary 16.2 (A target-relative obstruction to learning)**.** *For a Boolean target f, input distribution $\mathcal{D}$, and $0 < \epsilon < 1$, define*

$$J_{L,\epsilon}(f; \mathcal{D}) = \inf\{J_L(g) : \Pr_{\mathcal{D}}(g \neq f) < \epsilon\}.$$

Under the depth hypothesis of Theorem 16.1, if

$$J_{L,\epsilon}(f;\mathcal{D}) > U_{d,n,L},$$

then every sample of at most n inputs labeled by f yields width-first posterior Gibbs error with lower limit at least ϵ. Its correct-label probabilities $p_N(x)$ satisfy

$$\liminf_{N\to\infty} \Pr_{\mathcal{D}}\{p_N(x) < 2/3\} \geq \max\{0, (3\epsilon - 1)/2\}.$$

Proof. The rate ceiling excludes, asymptotically, every function of error less than ϵ. Every remaining function has error at least ϵ, proving the Gibbs assertion. If e_N is the displayed confidence-error probability, Gibbs error is at most $(1 - e_N)/3 + e_N$. Rearranging and taking lower limits proves the second claim. □

In particular, a polynomial sample guarantee would require suitable weak approximations whose normalized rate excess $L[J_L(g) - V_*]$ is only $O(\log^2(d + C + L))$, when the sample size is polynomial in dimension d, teacher complexity C, and depth L, and the statistical parameters are fixed. This is a necessary condition, not a proved approximation theorem for arbitrary small circuits. One should use a fixed error slack when passing from a Gibbs guarantee to a particular approximant.

The requirement is not vacuous merely because all rates converge to V_*. The uniform all-function upper bound gives normalized excess $O(d^2 + \log^2 L)$, whereas the ceiling for polynomially many examples has only squared-logarithmic dependence. A negative learning result would follow from a suitable approximation-excess lower bound for an explicit small target. A positive learning result must instead control inaccurate extensions within this much narrower interval. Neither conclusion follows from the common leading rate alone.

17 A finite-width upper bound at the limiting critical exponent

The scalar comparison yields a finite-width bound with an explicit subexponential prefactor. In this section the Gaussian scale is critical, $t = 1$, and the required clean output margin is $1/2$. No large-deviation limit is used. Write $H = L - 1$ and let $A_{N,f}$ denote the event $f(x)z_w(x) > 1/2$ on the entire Boolean cube.

Theorem 17.1 (Finite-width universal upper bound)**.** *There is a universal constant $C < \infty$ such that, for every $d, N, H \geq 1$ and every Boolean function f,*

$$\mu_{N,1}(A_{N,f}) \leq \min\left\{1, \exp\left[-NV_* + \frac{CN}{H} + CH\log(2N)\right]\right\}. \tag{8}$$

Here $V_* \in (1, 2)$ is the scalar limiting value defined above. The constant does not depend on input dimension, width, or depth. In fact, the same one-sided bound holds for $sz_w(x_0) > 1/2$ at any fixed input x_0 and sign $s \in \{-1, 1\}$. The full global-margin event therefore satisfies

$$\mu_{N,1}(A_N) \leq \min\left\{1, 2\exp\left[-NV_* + \frac{CN}{H} + CH\log(2N)\right]\right\}, \qquad A_N = \bigcup_f A_{N,f}.$$

We first record a modulus estimate for the scalar value. Extend V to zero by $V(0) = V_*$.

Lemma 17.2 (A uniform scalar-value modulus)**.** *For a universal C_V, every $0 \leq q \leq q' \leq 1$ satisfies*

$$0 \leq V(q) - V(q') \leq C_V\sqrt{q' - q}.$$

In addition, $0 \leq V_* - V(q) \leq C_V q$ for $0 \leq q \leq 1$.

Proof. The small-state bridge gives $V_* - V(q) \le 17q$ for $q \le 1/128$. For larger q, use $0 \le V(q) \le V_* < 2$ to obtain the second assertion with, for example, constant 256.

For $0 < q < q'$, we also have

$$V(q) - V(q') \le 5(q'/q - 1). \tag{9}$$

Indeed choose a path from q' arbitrarily close to optimal and of cost less than two; the preceding uniform scalar upper construction permits this. If it has at least one step, write η for its first preactivation control. Gaussian entropy duality at parameter $1/(4q')$ gives

$$\int z^2 \, d\eta(z) \le q' \left[4D(\eta \| \mathcal{N}(0, q')) + 2\log 2\right] < 10q'.$$

Use the same control and all the same subsequent states when starting at q. The first entropy changes by exactly

$$\frac{1}{2}\log(q/q') + \frac{1}{2}(1/q - 1/q') \int z^2 \, d\eta(z) \le 5(q'/q - 1).$$

If the selected path has zero steps, its cost is $1/(8q') < 2$; the zero-step cost from q is larger by at most $2(q'/q - 1)$. Taking the approximation error to zero proves (9).

Put $h = q' - q$. If $q \ge \sqrt{h}$, the relative modulus is at most $5\sqrt{h}$. Otherwise $q' \le \sqrt{h} + h \le 2\sqrt{h}$, and the second assertion gives $V(q) - V(q') \le V_* - V(q') \le 512\sqrt{h}$. The zero-width interval and $q = 0$ cases are immediate. □

Lemma 17.3 (Occupation with an additive prefix budget)**.** *Suppose $u_0, \dots, u_H \in (0, 1]$, nonnegative costs D_j, and $e \ge 0$ satisfy*

$$D_j \ge u_j - u_{j-1} + cu_{j-1}^2, \qquad \sum_{j=1}^{i} D_j \le C_0 u_i + e, \qquad \sum_{j=1}^{H} D_j \le 3.$$

For constants depending only on c, C_0, their minimum $a = \min_j u_j$ satisfies

$$H \le \frac{C_1}{a} + \frac{C_1 e}{a^2}, \qquad a \le \frac{C_2}{H} + C_2\sqrt{e/H}.$$

Proof. The reciprocal telescoping inequality gives, on any interval whose pre-states are at least b,

$$c\,\#\{\text{steps}\} \le b^{-2} \sum D_j + 1/u_{\text{end}} - 1/u_{\text{start}}.$$

If a exceeds the fixed threshold $1/256$, the total cost bound gives a universal length bound. Otherwise partition the path at its last exits from $[0, 2^k a]$, stopping when $2^k a \ge 1/256$. The prefix ending at the global minimum has cost at most $C_0 a + e$, so its length is at most $C/a + Ce/a^2$. In each subsequent dyadic block, remove its first transition. All remaining pre-states exceed $b = 2^k a$, the endpoint lies in $(b, 2b]$, and the block cost is at most $2C_0 b + e$. Its remaining length is therefore at most $C/b + Ce/b^2$. The final tail above $1/256$ has universal bounded length by the total cost bound, after again removing one transition. Summing the geometric series, and restoring the removed transitions, proves the first estimate. Solving its quadratic inequality for a gives the second. □

For $q > 0$ and $0 \le y < 1$, define the upper-tail scalar cost

$$I_+(q, y) = \inf \left\{ D(\eta \| \mathcal{N}(0, q)) : \int \tanh^2 z \, d\eta(z) \ge y \right\}.$$

It has the Chernoff dual formula

$$I_+(q,y) = \sup_{\lambda \geq 0} \left\{ \lambda y - \log \mathbb{E} \exp[\lambda \tanh^2(\sqrt{q}Z)] \right\}. \tag{10}$$

For y below the reference mean both sides are zero. Otherwise an exponential tilt with positive parameter realizes mean y; the tilted mean increases continuously from the reference mean to one. This proves the formula directly, without an asymptotic Cramer theorem. Two consequences will be used:

$$I_+(q,y) \geq V(q) - V(y), \tag{11}$$

$$I_+(q,y) \geq y - q + cq^2 \qquad (0 < q \leq 1), \tag{12}$$

where $c > 0$ is the strict scalar-potential constant. For the first, any feasible control has next state $q' \geq y$, so the definition and monotonicity of V give $V(q) \leq D(\eta \| \mathcal{N}(0,q)) + V(q') \leq D(\eta \| \mathcal{N}(0,q)) + V(y)$. For the second, entropy duality and $\log \mathbb{E} \exp[\tanh^2(\sqrt{q}Z)] \leq q - cq^2$ give the result. These inequalities also hold at $y = 0$ with the extension $V(0) = V_*$.

Proof of Theorem 17.1. After increasing C, the theorem is trivial if $N = 1$, $H = 1$, or $H > N$, since $V_* < 2$. Thus assume $N \geq 2$ and $2 \leq H \leq N$, and put $K = H - 1$. Fix one input x_0 and condition on its entire first hidden vector. Define

$$v_j = \frac{1}{N} \sum_{i=1}^{N} h_i^{j+1}(x_0)^2 \quad (0 \leq j \leq K).$$

All these numbers belong to $[0,1]$; the conditional initial value v_0 is arbitrary. If $v_0 = 0$, every later hidden vector at x_0 and its readout are zero, so the asserted bound is immediate. Otherwise, conditional on all preceding layers, the next preactivations at x_0 are independent Gaussians of variance v_{j-1}. Thus v_j is exactly an average of N independent $\tanh^2$ Gaussian variables, and, for every $y \geq 0$,

$$\Pr\{v_j \geq y \mid \text{previous layers}\} \leq \exp[-N I_+(v_{j-1}, y)].$$

When $v_{j-1} = 0$, the vector remains zero and the same statement has the evident degenerate interpretation. The right-hand-side rate is nonincreasing in its first argument: $\tanh^2(\sqrt{q}Z)$ increases pointwise with q, so its nonnegative exponential moments increase with q.

Take the grid mesh $h = N^{-4}$ and $J = N^4$ bins $[0,h], (h,2h], \ldots, ((J-1)h, 1]$. For a path of bin indices, let u_j be its upper endpoint and $l_j = u_j - h$ its lower endpoint, and put $u_0 = v_0$. By conditioning successively on full preceding layers, the probability of this bin path, together with $s z_w(x_0) > 1/2$ for any fixed sign s, is at most

$$\exp[-N\mathcal{A}(u)], \qquad \mathcal{A}(u) = \sum_{j=1}^{K} I_+(u_{j-1}, l_j) + \frac{1}{8u_K}.$$

For bins after the first, the actual event has $v_j > l_j$, which is contained in $v_j \geq l_j$; for the first bin the lower threshold is zero. The conditional readout variance is $v_K/N \leq u_K/N$, giving the last term. Monotonicity in the previous variance justifies replacing its actual value by u_{j-1} at every transition.

We bound the action of every grid path. If $\mathcal{A}(u) > V_*$, the claimed lower estimate below is automatic. Otherwise put $D_j = I_+(u_{j-1}, l_j)$ and $\tilde{D}_j = D_j + h$. By the modulus lemma, $V(l_i) - V(u_i) \leq C_V/N^2$ for every bin. Bellman's inequality on any suffix, together with $V(u_K) \leq 1/(8u_K)$, gives

$$\sum_{j=i+1}^{K} D_j + \frac{1}{8u_K} \geq V(u_i) - \frac{C_V K}{N^2}.$$

Consequently every prefix obeys

$$\sum_{j=1}^{i} \widetilde{D}_j \le \mathcal{A}(u) - V(u_i) + \frac{C_V K}{N^2} + Kh \le C_V u_i + e, \qquad e = \frac{C_V K}{N^2} + Kh.$$

Also $\sum_j \widetilde{D}_j \le V_* + Kh < 3$, and (12) yields the exact inequality

$$\widetilde{D}_j \ge u_j - u_{j-1} + c u_{j-1}^2.$$

Lemma 17.3 therefore supplies some $j \in \{0, \dots, K\}$ with

$$u_j \le \frac{C_1}{K} + C_1\sqrt{e/K} \le \frac{C_2}{K} + \frac{C_2}{N}.$$

The suffix bound at this layer gives

$$\mathcal{A}(u) \ge V(u_j) - \frac{C_V K}{N^2} \ge V_* - \frac{C_3}{K} - \frac{C_3}{N} - \frac{C_V K}{N^2}.$$

There are $J^K = N^{4K}$ bin paths. Summing their probability bounds, using $1/K \le 2/H$, and absorbing $C_3 + C_V K/N + 4K \log N$ into $CH \log(2N)$ proves the claimed one-sided bound conditionally on every first hidden vector with $v_0 > 0$. The proof used no restriction on v_0 beyond $v_0 \le 1$; the prefix inequality at $i = 0$ is automatic. Integrating this uniform bound over the first layer proves it unconditionally.

For a fixed target, $A_{N,f}$ is contained in $\{f(x_0) z_w(x_0) > 1/2\}$. The full global-margin event is contained in $\{|z_w(x_0)| > 1/2\}$, and the two possible signs cost only a factor of two. This proves both assertions. □

In particular, if $m = 2^d$, $N = m \exp(O(\sqrt{d \log d}))$, and $H = \exp(O(\sqrt{d \log d}))$ is chosen so that $N/H = o(m)$, then the correction $CN/H + CH \log(2N)$ is $o(m)$. This statement alone is an upper bound. In fact, the global-event version rules out matching lower masses with logarithmic error $o(m)$ uniformly over all Boolean functions: summing such lower bounds over their 2^m disjoint events contradicts the displayed upper bound. Section 18 makes this obstruction quantitative. A posterior comparison must retain the function-specific finite-width terms concealed by the common leading rate.

18 Why a common-exponent counting argument is insufficient

Let $m = 2^d$ and

$$E_{N,L} = C\left[\frac{N}{L-1} + (L-1)\log(2N)\right],$$

with C as in the finite-width universal upper bound. The target events $A_{N,f}$ are disjoint and their union is the global-margin event A_N. Therefore

$$\sum_f \mu_{N,1}(A_{N,f}) \le 2\exp(-NV_* + E_{N,L}).$$

This gives an immediate obstruction to transferring the common variational rate into uniform finite-width target masses.

Proposition 18.1 (Unavoidable function-specific logarithmic overhead)**.** *If a number R satisfies*

$$\mu_{N,1}(A_{N,f}) \geq \exp(-NV_* - R) \qquad \text{for every Boolean } f,$$

then necessarily

$$R \geq m \log 2 - \log 2 - E_{N,L}.$$

In particular, in a regime where $E_{N,L} = o(m)$, a uniform lower bound with $R = o(m)$ is impossible.

Proof. Sum the asserted lower bound over the 2^m sign patterns and compare with the displayed upper bound on their union. □

More generally, for every $r \in \mathbb{R}$, the number of functions whose target event has probability at least e^{-NV_*-r} is at most $2e^{E_{N,L}+r}$. Thus the common limiting exponent must conceal substantial function-specific finite-width terms. Their presence is not an artifact of a particular coupling or concentration proof.

This observation does not establish good learning: the missing terms could favor simple targets, or could resemble a nearly uniform prior over truth tables. It also does not establish bad learning. A posterior sample lower bound requires a target-relative comparison that retains these terms; a shared leading exponent alone cannot supply one.

19 An exact symmetry useful for comparing targets

The first-layer bias has the same Gaussian variance as each input coefficient. This gives a slightly larger symmetry than coordinate permutations of the Boolean cube. The result below holds at finite width and every prior scale, not only for the critical rate.

Proposition 19.1 (Projective coordinate symmetry)**.** *Write $y = (1, x_1, \ldots, x_d)$ and let π be any permutation of $\{0, \ldots, d\}$. Define*

$$a_\pi(x) = y_{\pi(0)}, \qquad T_\pi(x)_j = y_{\pi(j)}/y_{\pi(0)} \quad (1 \leq j \leq d).$$

The map T_π is a bijection of the Boolean cube. If z is a network drawn from the Gaussian prior, then

$$x \longmapsto a_\pi(x) z(T_\pi(x))$$

has exactly the same law as z. Consequently, writing $f^\pi(x) = a_\pi(x) f(T_\pi(x))$, the target events $A_{N,f}$ and A_{N,f^π} have equal prior probabilities. At critical scaling, $J_L(f) = J_L(f^\pi)$.

In particular, for every m for which the indicated degrees exist, all parity functions of degree $2m$ or $2m+1$ have the same prior probability and the same critical rate. Constants and dictators are the first such pair of degree classes.

Proof. Permute the $d+1$ columns of the first-layer matrix by π. Its Gaussian law is unchanged. The permuted input vector satisfies

$$(y_{\pi(0)}, \ldots, y_{\pi(d)}) = a_\pi(x)(1, T_\pi(x)).$$

Every subsequent hidden activation is odd, and there are no subsequent biases. Thus the factor $a_\pi(x) \in \{-1, 1\}$ passes through all hidden layers and the linear readout. This proves the asserted identity in law. The induced transformation of projective sign vectors is invertible, which proves that T_π is a bijection. The equality of target-event probabilities follows by changing variables on

the cube; the rate equality follows either from these exact finite-width identities or by the same transformation in the variational functional.

For the final assertion it suffices to swap coordinate zero with an input coordinate i. If a parity has even degree and does not involve i, the transformation adds i. If a parity has odd degree and does involve i, it removes i. In the other two cases it leaves the parity unchanged. Ordinary input permutations identify all subsets of a given size. □

For uniform input sampling, transforming each example by T_π and multiplying its label by a_π also identifies the corresponding posterior learning problems, including Gibbs error and any fixed posterior-confidence threshold. This exact symmetry does not identify all parity degrees, and does not by itself give a sample-complexity lower bound.

20 A necessary depth for bounded absolute rate

The common limiting rate does not mean that a bounded rate is attained at every depth. Gaussian entropy transport gives a complementary constraint. This is a statement about representation rates, not a sample-complexity lower bound.

For $y = (y_0, \dots, y_d) \in \{-1, 1\}^{d+1}$ define the odd extension

$$\widetilde{f}(y) = y_0 f(y_1/y_0, \dots, y_d/y_0),$$

and let $k = k_{\mathrm{aug}}(f)$ be its number of relevant coordinates. Here a coordinate is relevant when some edge parallel to that coordinate has opposite endpoint labels. Since $\widetilde{f}$ is odd, $1 \le k \le d+1$. The count is unchanged by permutations of the augmented coordinates. In particular, constants and dictators both have $k = 1$. A parity of original degree j has $k = j$ when j is odd and $k = j + 1$ when j is even.

Theorem 20.1 (Entropy transport and depth)**.** *Let $J = J_L(f)$ and $h = L - 2$. Put*

$$B_h(J) = \begin{cases} (1 + \sqrt{2J/h})^{2h}, & h \ge 1, \\ 1, & h = 0. \end{cases}$$

Then

$$k_{\mathrm{aug}}(f) \le 64 J B_h(J), \tag{13}$$

$$d \le 8J\big(1 + \sqrt{2J/k_{\mathrm{aug}}(f)}\big)^2 B_h(J). \tag{14}$$

In particular $B_h(J) \le \exp(2\sqrt{2hJ})$. If $J_L(f) \le K$ and $d > 8K(1 + \sqrt{2K})^2$, then

$$L \ge 2 + \frac{1}{8K}\left[\log \frac{d}{8K(1 + \sqrt{2K})^2}\right]^2.$$

Thus every target, including a constant, requires depth $\Omega((\log d)^2)$ to have a uniformly bounded critical rate as $d \to \infty$.

We use the Gaussian transport inequality in its correlated form. If $\eta \ll \mathcal{N}(0, Q)$ has entropy D, there is a coupling (Z, G) with these two marginals such that

$$\mathbb{E}\|Q^{+1/2}(Z - G)\|^2 \le 2D. \tag{15}$$

Both variables lie in $\operatorname{range} Q$. This is the standard Gaussian quadratic transport inequality [4], applied after whitening on that range; singularity causes no change. In particular one common coupling works for all input coordinates, rather than separate coordinatewise couplings.

Proof of Theorem 20.1. Take any admissible variational path of total cost $A = r + \sum_{s=1}^{L-1} D_s < \infty$, where r is its readout cost. Extend every feature oddly to the augmented cube as above. This is an injective linear transformation of the feature vector and preserves relative entropy. The Gaussian reference and the tanh map commute with this extension. Finite entropy ensures that every feature coordinate is in $(-1, 1)$, so coordinatewise atanh lifts each controlled feature law to its preactivation law with exactly the same entropy.

For a later hidden layer with input second moment Q, apply (15). For any two augmented inputs y, y', put $b = e_y - e_{y'}$. Cauchy–Schwarz on range Q and Minkowski give

$$\|b^T Z\|_{L^2} \le \|b^T G\|_{L^2} + \|b^T (Z - G)\|_{L^2} \le (1 + \sqrt{2D_s})\sqrt{b^T Q b}.$$

Since tanh is 1-Lipschitz, the L^2 distance of the two output features is at most this bound. Hence each later layer multiplies pairwise feature distances by at most $1 + \sqrt{2D_s}$. Write

$$P = \prod_{s=2}^{L-1} (1 + \sqrt{2D_s}),$$

with the empty product equal to one.

At the first layer, the field $y \mapsto W \cdot y$ identifies W uniquely. Its controlled law has entropy D_1 relative to $\mathcal{N}(0, I_{d+1}/d)$. The Gaussian transport coupling therefore satisfies $\mathbb{E}\|W - G\|^2 \le 2D_1/d$. If I is the set of the k relevant augmented coordinates, projection and Minkowski give

$$\sum_{i \in I} \mathbb{E} W_i^2 \le \frac{1}{d}\left(\sqrt{k} + \sqrt{2D_1}\right)^2, \qquad \mathbb{E}\|W\|^2 \le \frac{1}{d}\left(\sqrt{d+1} + \sqrt{2D_1}\right)^2. \tag{16}$$

A first-layer edge in direction i has feature distance at most $2\sqrt{\mathbb{E} W_i^2}$.

The readout admits a representation $z(y) = \mathbb{E}[a(H)H(y)]$ with $\mathbb{E} a(H)^2 = 2r$: one may take $a(H) = z^T Q^+ H$. Each relevant direction has a bichromatic edge, and the output margin makes the output difference on that edge at least one. Consequently

$$1 \le 2\sqrt{2r}\, P \sqrt{\mathbb{E} W_i^2} \qquad (i \in I).$$

Summing the squared inequalities and applying the first estimate in (16) proves the sharper budget form

$$d \le 8 r P^2 \left(1 + \sqrt{2D_1/k}\right)^2. \tag{17}$$

Using the full trace instead gives

$$k \le \frac{8 r P^2}{d}\left(\sqrt{d+1} + \sqrt{2D_1}\right)^2.$$

If $A < d$, its right side is at most $64 A P^2$. If $A \ge d$, the same conclusion follows from $k \le d + 1 \le 2A$.

The function $u \mapsto \log(1 + \sqrt{2u})$ is concave on $[0, \infty)$. For $h \ge 1$, Jensen's inequality and $\sum_{s=2}^{L-1} D_s \le A$ give

$$P^2 \le \left(1 + \sqrt{2A/h}\right)^{2h} \le e^{2\sqrt{2hA}}.$$

The same assertion with $P = 1$ holds for $h = 0$. Substituting $r, D_1 \le A$ in the preceding bounds and taking $A \downarrow J$ proves both inequalities. The final depth bound follows by taking logarithms of (14) and using $k \ge 1$. □

Remark 20.2 (What the bound does and does not control)**.** *For fixed rate budget K, (13) also requires depth $\Omega((\log k)^2)$ when $k \to \infty$. Any Boolean predictor that has uniform-distribution error strictly less than $1/2$ on a parity of j specified coordinates must depend on each of those coordinates. Thus the same relevant-coordinate restriction applies to every such predictor. Neither statement compares posterior masses of functions consistent with a training sample. A bound on pairwise feature distances also does not by itself bound the number of possible correlated feature laws, so it cannot be read as a learning theorem. The ambient-dimension restriction above is compatible with Theorem 2.1, whose sufficient depth is polynomial in d.*

21 A target-relative separation at every fixed depth

The transport bound also gives a comparison beyond depth two. The comparison is asymptotic in the input dimension and does not order all targets at every finite dimension or at arbitrary growing depth.

Theorem 21.1 (Fixed-depth critical separation)**.** *Let $L \geq 2$, $d \geq 1$, and let k be the number of relevant coordinates of the odd augmented extension of a Boolean target f. Then*

$$J_L(f) \geq \frac{1}{64} \min\{d^{1/(L-1)}, (dk)^{1/L}\}. \tag{18}$$

For the constant target, and hence for a signed dictator by the projective coordinate symmetry,

$$\frac{1}{64} d^{1/L} \leq J_L(\mathbf{1}) \leq 5000 L\, d^{1/L}. \tag{19}$$

Consequently

$$\frac{J_L(f)}{J_L(\mathbf{1})} \geq \frac{1}{320000L} \min\{d^{1/[L(L-1)]}, k^{1/L}\}.$$

For every fixed L, a sequence of targets with $k \to \infty$ therefore has a rate ratio tending to infinity as $d \to \infty$. In particular, every growing-degree parity eventually has strictly larger rate than a constant at each fixed depth. For full parity this strict comparison also holds whenever $L = o(\sqrt{\log d / \log\log d})$ and $d \to \infty$.

Proof. Put $J = J_L(f)$ and $h = L - 2$. Every admissible path has readout cost at least $1/8$: a single output has magnitude at least $1/2$, whereas each final feature has squared magnitude at most one. Thus $J \geq 1/8$. The transport estimate gives

$$d \leq 8 J B_h(J) (1 + \sqrt{2J/k})^2, \qquad B_0(J) = 1, \quad B_h(J) = (1 + \sqrt{2J/h})^{2h} \ (h \geq 1).$$

For $h \geq 1$, $2 + 4J/h \leq 20J$, so $B_h(J) \leq (20J)^h$. The same bound holds for $h = 0$. Also $(1 + \sqrt{2J/k})^2 \leq 2 + 4J/k$. Consequently

$$d \leq 16 \cdot 20^h J^{h+1} + \frac{32 \cdot 20^h}{k} J^{h+2}.$$

At least one summand is at least $d/2$. This proves

$$J \geq \min\left\{ \left(\frac{d}{32 \cdot 20^h}\right)^{1/(h+1)}, \left(\frac{dk}{64 \cdot 20^h}\right)^{1/(h+2)} \right\},$$

which implies (18) with the stated universal constant.

For completeness we construct the constant-target upper bound directly in the exact variational functional. Write $c = 1/100$. For $0 \le q \le 2$ and $0 \le \mu \le 1$, put

$$\psi_q(\mu) = \mathbb{E}\tanh(\mu + \sqrt{q}Z), \qquad Z \sim N(0,1).$$

The function is continuous and strictly increasing, vanishes at zero, and satisfies $\psi_q(\mu) \ge c\mu$. Indeed its derivative is at least $\Pr(|Z| \le 1)\operatorname{sech}^2(1+\sqrt{2}) > 1/100$ throughout this parameter range. Thus every $m \in [0, c]$ is realized by some $\mu \in [0, 1]$ with $\mu \le m/c$.

Prescribe the common hidden-feature means

$$m_s = c\, d^{-(L-s)/(2L)}, \qquad 1 \le s \le L-1.$$

At the first layer shift only the Gaussian bias by μ_1, chosen so that $\psi_{(d+1)/d}(\mu_1) = m_1$. Its relative entropy is $d\mu_1^2/2 \le d^{1/L}/2$. The feature law is invariant under input sign translations, has mean $m_1\mathbf{1}$, and hence its second moment satisfies $Q_1 \succeq m_1^2\mathbf{1}\mathbf{1}^T$.

Inductively, suppose the preceding feature law is translation invariant and has mean $m_{s-1}\mathbf{1}$. Its covariance in the Gaussian transition is its second moment Q_{s-1}, whose diagonal is a common value $q_{s-1} \le 1$. Replace the centered Gaussian preactivation by $N(\mu_s\mathbf{1}, Q_{s-1})$, choosing $\psi_{q_{s-1}}(\mu_s) = m_s$. The mean direction belongs to the range of Q_{s-1}, and the rank-one second-moment bound gives the entropy estimate

$$\frac{\mu_s^2}{2}\mathbf{1}^T Q_{s-1}^{+}\mathbf{1} \le \frac{\mu_s^2}{2m_{s-1}^2} \le \frac{1}{2c^2}d^{1/L}.$$

Coordinatewise tanh is injective, so its pushforward has the same relative entropy from the transition reference. Translation invariance and the prescribed common mean are preserved. This proves the induction, including singular second-moment matrices.

Finally choose the output $z = \mathbf{1}/2$. Since $Q_{L-1} \succeq m_{L-1}^2\mathbf{1}\mathbf{1}^T$, its readout cost is at most

$$\frac{1}{8}\mathbf{1}^T Q_{L-1}^{+}\mathbf{1} \le \frac{1}{8m_{L-1}^2} = 1250d^{1/L}.$$

Summing the first-layer cost, the $L-2$ later hidden costs, and the readout cost proves the upper bound in (19). Its lower bound is (18) with $k = 1$. The ratio estimate follows immediately. For full parity $k \ge d$, the logarithm of its ratio lower bound is $\log d/[L(L-1)] - \log(320000L)$, which tends to infinity in the stated growing-depth regime. □

This theorem compares unconditioned target rates. It does not compare the best training-consistent competitors or prove a posterior sample complexity lower bound. The estimates deliberately leave the regime of large growing depth open; universal rate flattening is compatible with them.

22 The absolute-rate scale at every depth

The necessary depth from entropy transport and the constant-target constructions fit into a single rate scale. This identifies the order of the smallest absolute rate, up to universal multiplicative constants; it does not assert that constants are exact minimizers at every depth.

Theorem 22.1 (A uniform depth–dimension rate profile)**.** *Define*

$$\Phi_L(d) = 1 + L\big(d^{1/(2L)} - 1\big)^2, \qquad d \ge 1, \quad L \ge 2.$$

There are universal constants $0 < c < C < \infty$ such that every Boolean function satisfies

$$J_L(f) \geq c\Phi_L(d),$$

whereas constants and dictators satisfy

$$J_L(f) \leq C\Phi_L(d).$$

Thus the minimum critical absolute rate over all Boolean targets, and the rate of a constant target, are both comparable to $\Phi_L(d)$ uniformly in dimension and depth. For $L \geq \log(d+1)$, this scale is comparable to $1 + [\log(d+1)]^2/L$. These statements concern prior rates, rather than sample complexity or the ordering of every pair of targets.

Proof. Take an admissible path of cost $A = r + \sum_{s=1}^{L-1} D_s$. The exact projected-trace budget (17), with $k \geq 1$, gives

$$d \leq 8r \prod_{s=1}^{L-1} (1 + \sqrt{2D_s})^2 \leq 4(1 + \sqrt{2r})^2 \prod_{s=1}^{L-1} (1 + \sqrt{2D_s})^2.$$

Concavity of $u \mapsto \log(1 + \sqrt{2u})$ and Jensen's inequality on these L costs imply

$$d \leq 4(1 + \sqrt{2A/L})^{2L}.$$

Taking $A \downarrow J_L(f)$ yields the explicit lower bound

$$J_L(f) \geq \frac{L}{2} \left[(d/4)^{1/(2L)} - 1 \right]_+^2. \tag{20}$$

Also $J_L(f) \geq 1/8$ by the single-input readout bound.

For $d \geq 16$, write $u = \log d/(2L)$ and $v = \log 4/(2L)$. Then $u \geq 2v$ and $e^v \leq \sqrt{2}$, so

$$e^u - 1 = e^v(e^{u-v} - 1) + (e^v - 1) \leq (\sqrt{2} + 1)(e^{u-v} - 1).$$

Combining this with (20) and the readout bound proves $J_L(f) \geq c\Phi_L(d)$ for a universal $c > 0$. For $1 \leq d < 16$, the function $L \mapsto L(d^{1/(2L)} - 1)^2$ is nonincreasing on $(0, \infty)$: putting $t = \log d/(2L)$ reduces this to monotonicity of $(e^t - 1)^2/t$. Its derivative is positive for $t > 0$, since $2te^t - (e^t - 1) > 0$. Hence $\Phi_L(d) \leq 3$ for $L \geq 2$, and the readout bound proves the same lower comparison.

For the upper comparison, the constant-target seed satisfies $S_{\mathbf{1}} \leq C_0 \log(d+1)$. Theorem 11.1 therefore gives, whenever $L \geq C_1 \log(d+1)$,

$$J_L(\mathbf{1}) \leq V_* + C_2 \frac{[\log(d+1)]^2 + [\log(2L)]^2}{L} \leq C_3 \left(1 + \frac{[\log(d+1)]^2}{L} \right).$$

Here $[\log(2L)]^2/L$ is universally bounded for $L \geq 2$. For $d \geq 2$, $\log(d+1) \leq 2 \log d$ and $e^t - 1 \geq t$ show that this last expression is at most $C_4\Phi_L(d)$. For $d = 1$ it is bounded universally and $\Phi_L(1) = 1$.

In the remaining range $L < C_1 \log(d+1)$, use (19): $J_L(\mathbf{1}) \leq 5000Ld^{1/L}$. For $d \geq 2$, the parameter $t = \log d/(2L)$ obeys $t \geq 1/(4C_1)$, so

$$\frac{e^{2t}}{(e^t - 1)^2} = (1 - e^{-t})^{-2} \leq (1 - e^{-1/(4C_1)})^{-2}.$$

Thus this upper bound is at most a universal multiple of $L(d^{1/(2L)} - 1)^2$. When $d = 1$ the remaining range has universally bounded L, and the same conclusion with $\Phi_L(1) = 1$ follows. The dictator case follows from projective symmetry.

Finally, when $L \geq \log(d+1)$, $t = \log d/(2L) \in [0, 1/2]$, so $t \leq e^t - 1 \leq e^{1/2}t$; replacing $\log d$ by $\log(d+1)$ changes $1 + (\log d)^2/L$ by at most universal factors. This proves the final scale comparison. □

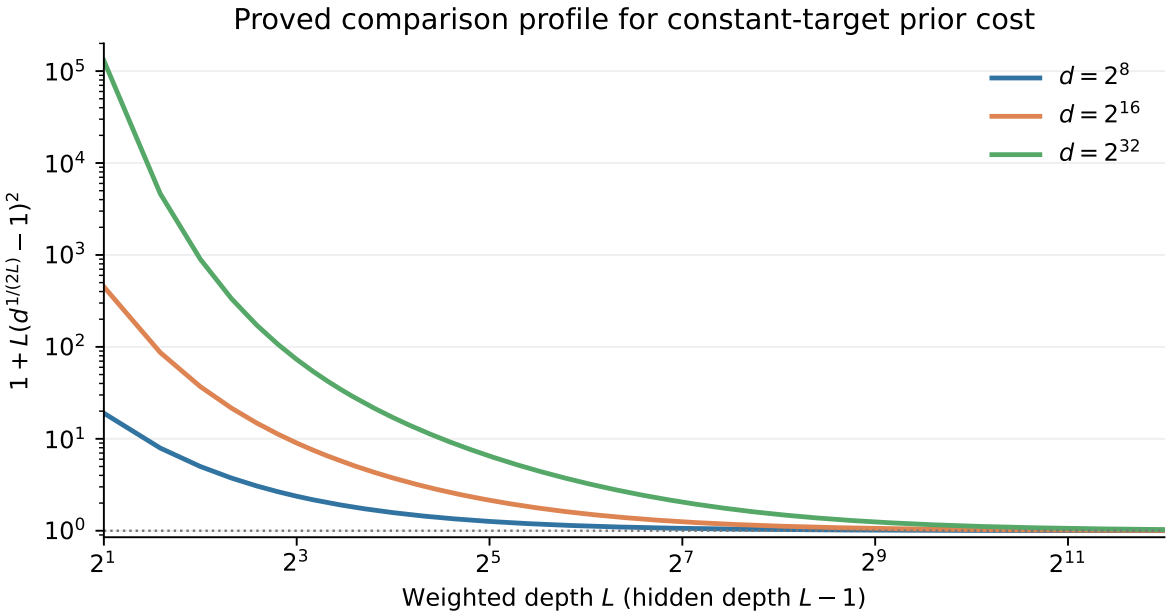


Figure 1: The comparison function in the all-depth constant-target theorem. The actual critical rate lies between universal constant multiples of this function; the curves are not numerical estimates of that rate. The shallow-depth growth and the transition to bounded cost are parts of one quantitative statement. Depth counts weighted layers.

23 A strict target-relative ordering at depth two

The common limiting rate does not imply equality at finite depth. At depth two one can determine a strict comparison between simple targets and every nontrivial parity.

Theorem 23.1 (Constants have lower critical rate than nontrivial parities)**.** *At critical scaling and depth $L = 2$, let $\mathbf{1}$ denote the constant-positive Boolean function. If f is nonconstant and antipodally even, meaning $f(-x) = f(x)$, then*

$$J_2(f) > J_2(\mathbf{1}).$$

Consequently every parity of degree at least two has strictly larger critical rate than a constant or a dictator. The negative constant and signed dictators have the same rate as $\mathbf{1}$.

Write

$$X_f(W) = \frac{1}{m} \sum_{x \in \{-1,1\}^d} f(x) \tanh(W \cdot \bar{x}), \qquad m = 2^d.$$

The proof uses the following exact formula for character targets.

Lemma 23.2 (Depth-two character rate)**.** *If f is a Boolean character, including the constant character, then*

$$J_2(f) = \max_{t \geq 0} \left\{ \frac{t}{2} - \log \mathbb{E}_W \exp\left(\frac{t^2 X_f(W)^2}{2} \right) \right\}.$$

The maximizing parameter for the constant character exists and is strictly positive. For an arbitrary Boolean f, the right-hand side is a lower bound for $J_2(f)$.

Proof. At depth two, the output vector has the distribution

$$z_N = \frac{1}{N}\sum_{i=1}^{N} Z_i h(W_i), \qquad h(W)(x) = \tanh(W \cdot \bar{x}),$$

where $Z_i \sim \mathcal{N}(0,1)$ are independent of the first-layer weights. Its logarithmic moment generating function is

$$\Lambda(\theta) = \log \mathbb{E}_W \exp\{(\theta \cdot h(W))^2/2\}.$$

It is finite for every θ. The depth-two output rate is the convex Legendre transform $I(z) = \sup_\theta(\theta \cdot z - \Lambda(\theta))$. This also follows directly from the variational functional: among joint laws of (h, Z) having a given marginal ν and $\mathbb{E}[Zh] = z$, the minimum relative entropy in the Gaussian Z coordinate is $z^T Q(\nu)^+ z/2$. Minimizing the remaining marginal entropy gives precisely the entropy formula for I.

For completeness the rate is finite and continuous on the whole output space. The first feature covariance K is positive definite. Jensen's inequality gives $\Lambda(\theta) \geq \mathbb{E}(\theta \cdot h)^2/2 \geq \lambda_{\min}(K)\|\theta\|_2^2/2$. Thus its dual is finite; the Gaussian scalar also makes every output direction available. Equivalently, the finite-dimensional entropy duality here has no boundary obstruction. Strict or non-strict output margins consequently give the same infimum.

The law of the output vector is invariant under input translations $x \mapsto sx$ and under global output negation. For a character f, the transformation $z(x) \mapsto f(s)z(sx)$ preserves both this law and the constraints $f(x)z(x) \geq 1/2$. Averaging over all s and using convexity replaces any feasible z by af, where $a = m^{-1}\sum_x f(x)z(x) \geq 1/2$, without increasing I. Convexity and $I(0) = 0$ then reduce the optimum to $z = f/2$. The same group averaging in the dual restricts θ to tf/m, giving the displayed formula. Symmetry in t permits $t \geq 0$. For general f, substituting this dual vector directly gives the asserted lower bound, because $\theta \cdot z \geq t/2$ on the target-margin event.

For the constant character, $X_{\mathbf{1}}$ is nonzero with positive probability. Therefore for some $a, p > 0$, $\Pr\{|X_{\mathbf{1}}| \geq a\} \geq p$, and the objective tends to $-\infty$ as $t \to \infty$. Its derivative at zero is $1/2$. It has a finite, strictly positive maximizer. □

Proof of Theorem 23.1. Write $W \cdot \bar{x} = b + w \cdot x$. For every real u,

$$\tanh(b+u) + \tanh(b-u) = \frac{\sinh(2b)}{\cosh(b+u)\cosh(b-u)}.$$

For $b \neq 0$ every such antipodal pair sum is nonzero and has the same sign as b. If f is even and nonconstant, it takes both signs on the set of antipodal pairs. Strict triangle inequality therefore gives, pointwise for every finite w and $b \neq 0$,

$$|X_f(b,w)| < |X_{\mathbf{1}}(b,w)|.$$

The Gaussian bias is nonzero almost surely, so for every $t > 0$,

$$\mathbb{E}e^{t^2 X_f^2/2} < \mathbb{E}e^{t^2 X_{\mathbf{1}}^2/2}.$$

Evaluate the lower bound in the preceding lemma at a positive maximizer for the constant character. This proves the strict rate inequality.

Every positive even-degree parity is antipodally even and nonconstant. A parity of odd degree at least three has the same rate as a parity of the preceding even degree by the projective coordinate symmetry. That symmetry also identifies constants and dictators, and global output negation identifies their negatives. □

The strict gap implies an exponentially small zero-data prior ratio between such a target and a constant as width tends to infinity at this fixed depth. It does not give a sample lower bound: a few oppositely labeled observations already eliminate the constant competitor. The exact ordering at every finite dimension and arbitrary depth remains open; the preceding comparison covers growing-relevance targets at fixed depth and sufficiently large dimension. Neither result completely orders training-consistent extensions. The later sample lower bound uses a separate comparison with an explicit interpolating function.

The pointwise comparison used in the proof is specific to a first-layer feature. For example, on two input bits put

$$h_1 = \tanh(x_1 + x_2 - 1), \quad h_2 = \tanh(x_1 + x_2 + 1), \quad h_3 = \tanh(1), \qquad g = \tanh(h_1 - h_2 + h_3).$$

This uses only first-layer biases. The feature g is positive when $x_1 x_2 = 1$ and negative when $x_1 x_2 = -1$: its two values are $\tanh(2\tanh 1 - \tanh 3) > 0$ and $-\tanh(\tanh 1) < 0$. Consequently its parity coefficient has greater magnitude than its constant coefficient. Thus the preceding pointwise domination fails already for a second-hidden-layer feature. This example does not reverse the full prior rate comparison; it identifies a limitation of this proof method.

24 A polynomial sample lower bound at each fixed critical depth

A low-rate interpolating halfspace gives a sample lower bound for the actual critical posterior. The bound is polynomial in the parity and ambient dimensions, and the width limit is taken before those dimensions grow.

Lemma 24.1 (Amplifying a first-feature mean through its second moment)**.** *Let μ_1 be the uncontrolled first-feature law. Suppose a feature law ν satisfies $D(\nu\|\mu_1) \le D$, where $D \ge 1$, and its mean h satisfies $g(x)h(x) \ge a$ at every input, with $0 < a \le 1$ and g Boolean. Set $R = D/a^2$ and $q = R^{1/L}$. If $q \le D$, then*

$$J_L(g) \le 5000 L R^{1/L}.$$

Proof. Put $\theta = q/D \le 1$. The first controlled law $(1-\theta)\mu_1 + \theta\nu$ has entropy at most $\theta D = q$ by convexity, and its second moment dominates $\theta h h^T$. Thus it has a tracked rank-one vector $b_1 = \sqrt{\theta} h$ with

$$g(x) b_1(x) \ge \sqrt{\theta} a = q^{-(L-1)/2}.$$

This is a second-moment component; it need not equal the first controlled law's mean.

For $L = 2$, choose the readout $z = h/(2a)$. Its cost is at most $1/(8\theta a^2) = q/8$, proving the assertion in this case. Suppose $L \ge 3$ and put $c = 1/100$. For every $0 \le v \le 1$ and $u \ge 0$, the elementary estimate used in the constant-target construction gives

$$\mathbb{E}\tanh(u + \sqrt{v} Z) \ge c \min\{u, 1\}.$$

At the second hidden layer shift the reference Gaussian by $\sqrt{q}\, b_1$. Its mean-shift entropy is at most $q/2$, because $Q_1 \succeq b_1 b_1^T$. After tanh, its actual mean b_2 satisfies $g(x) b_2(x) \ge c q^{-(L-2)/2}$. Its second moment dominates $b_2 b_2^T$.

At each hidden layer $s = 3, \dots, L-1$, shift the reference Gaussian by $(\sqrt{q}/c) b_{s-1}$. The entropy is at most $q/(2c^2)$, and the new feature mean satisfies

$$g(x) b_s(x) \ge c q^{-(L-s)/2}.$$

The induction uses only that every reference diagonal is at most one, not that the diagonals or feature magnitudes are equal. If some shift coordinate exceeds one in magnitude, the displayed minimum estimate still gives the required lower bound. All shift vectors lie in the corresponding covariance range; the formulas therefore also apply to singular covariances. Tanh pushforward cannot increase the entropy.

Finally take $z = b_{L-1}/(2cq^{-1/2})$. It has the required global margin and Gaussian readout cost at most $q/(8c^2)$. Summing all costs gives the claimed bound. □

Lemma 24.2 (A low-rate majority interpolant for parity samples)**.** *Let the labels be a parity on a nonempty coordinate set S, and draw $n \geq 1$ independent uniform cube inputs. Fix one $i_0 \in S$ and an odd number $m \leq d-1$ of other coordinate indices I. There is a sample-dependent Boolean majority function*

$$g(x) = \operatorname{sign}\left(\sum_{j\in I} s_j x_j\right), \qquad s_j \in \{-1, 1\},$$

which fits every training label with probability at least $1 - n\exp[-m/(6n)]$. For every sample, this function admits a first-feature law ν with

$$D(\nu\|\mu_1) \leq dm, \qquad g(x)\mathbb{E}_\nu H(x) \geq 1/100 \quad \textit{for every input } x.$$

Consequently, whenever $dm \geq 10000$,

$$J_L(g) \leq 5000L(10000dm)^{1/L} \qquad (L \geq 2).$$

Proof. Write Y_i for the parity label of training row i, and choose

$$s_j = \operatorname{sign}\left(\sum_{i=1}^{n} Y_i X_{ij}\right),$$

with either fixed convention for a zero sum. Conditional on the entire label vector Y, all coordinate columns other than i_0 are independent uniform sign vectors: the missing parity coordinate is uniquely determined by these columns and Y. Thus $A_{ij} = Y_i X_{ij}$, for $j \in I$, are independent uniform sign columns conditionally on Y.

If $T = \sum_{i=1}^{n} \varepsilon_i$ is a sum of independent uniform signs, exchangeability gives

$$\mathbb{E}\left[\varepsilon_i \operatorname{sign}(T)\right] = \frac{\mathbb{E}|T|}{n} \geq \frac{1}{\sqrt{3n}}.$$

Indeed $\mathbb{E}T^2 = n$ and $\mathbb{E}T^4 \leq 3n^2$, while Holder's inequality gives $\mathbb{E}T^2 \leq (\mathbb{E}|T|)^{2/3}(\mathbb{E}T^4)^{1/3}$. For each training row, the variables $s_j A_{ij}$ are therefore independent across j, bounded by one in absolute value, and have mean at least $1/\sqrt{3n}$. Hoeffding's inequality bounds the probability that their sum is nonpositive by $e^{-m/(6n)}$. A union bound proves the training-fit probability. Dependence between different training rows is irrelevant to that union bound.

Since m is odd, the affine form defining g is a nonzero integer at every cube input, so its global signed margin is at least one. Shift the first-row Gaussian weight mean to have coefficient s_j on I and zero elsewhere, including its bias. This costs exactly $dm/2$ in relative entropy. At each input its residual Gaussian variance is $(d+1)/d \leq 2$, and its feature mean is

$$\mathbb{E}\tanh\left(\sum_{j\in I} s_j x_j + \sqrt{(d+1)/d}\, Z\right).$$

The elementary mean-response bound from the constant-target construction gives signed mean at least $1/100$ everywhere. Data processing therefore proves the first-feature-law assertion. Apply Lemma 24.1 with $D = dm$ and $a = 1/100$. Its hypothesis $(10000D)^{1/L} \le D$ holds whenever $D \ge 10000$ and $L \ge 2$. □

Theorem 24.3 (Critical parity sample lower bound)**.** *Fix $L \ge 2$ and $0 < \eta < 1$. Let the target be parity on $k \le d$ specified input coordinates, and put*

$$M = \min\{k, d^{1/(L-1)}\}.$$

Suppose $M \to \infty$ and $1 \le n \le M^{1-\eta}$. Draw n independent uniform cube inputs labeled by this parity. For all sufficiently large M, with training-sample probability at least $1 - 1/(2M)$, the globally margin-conditioned posterior at $\gamma = 1$ has Gibbs error tending to $1/2$ as $N \to \infty$ at this fixed d, L, n. If $p_N(x)$ is its posterior probability of the correct label, then on the same event

$$\liminf_{N\to\infty} \Pr_x\{p_N(x) < 2/3\} \ge 1/4.$$

For full parity, this gives $n \le d^{(1-\eta)/(L-1)}$. If $d \ge k^{L-1}$, it gives $n \le k^{1-\eta}$.

Proof. Let m be the smallest odd integer at least $12n\log(2M(n+1))$. Since $n \le M^{1-\eta}$, one has $m = O(n\log(M+1)) = o(M) \le o(d)$, so $m \le d-1$ for large M. Lemma 24.2 supplies, with probability at least $1 - 1/(2M)$, a training-consistent halfspace and a first-feature law of entropy at most $D = dm$ with mean margin $a = 1/100$.

Apply Lemma 24.1. Its hypothesis $(D/a^2)^{1/L} \le D$ is equivalent to $D^{L-1} \ge 10000$ and holds for large M. The resulting training-consistent competitor has rate at most

$$5000L(10000dm)^{1/L}.$$

Any Boolean function depending on every one of the k parity coordinates has at least k relevant augmented coordinates. By Theorem 21.1, its critical rate is at least

$$\frac{1}{64}\min\{d^{1/(L-1)}, (dk)^{1/L}\} = \frac{1}{64}(dM)^{1/L}.$$

The ratio of the preceding competitor upper bound to this lower bound is at most

$$CL\,M^{-\eta/L}[\log(M+1)]^{1/L},$$

which tends to zero. Thus there is a positive rate gap between a training-consistent competitor and every function retaining all parity coordinates.

The exact finite-domain posterior-rate principle now implies that posterior mass on the latter functions tends to zero in the width limit. A function omitting any one parity coordinate has uniform parity error exactly $1/2$, even if it depends on irrelevant coordinates. This proves the Gibbs limit. Finally $\mathbb{E}_x p_N(x) \to 1/2$ and Markov's inequality give $\Pr_x\{p_N(x) < 2/3\} \ge 1 - (3/2)\mathbb{E}_x p_N(x)$. □

The theorem concerns the actual posterior, but takes width to infinity at fixed dimension, depth, and sample size. It gives no finite-width threshold uniform in growing dimension and no corresponding lower bound for posterior-majority prediction at confidence threshold $1/2$.

Corollary 24.4 (Slowly growing learner depth)**.** *Fix $0 < \eta < 1$. For full parity, the same conclusion holds at sample sizes $n \le d^{(1-\eta)/(L-1)}$ along integer depths $L = L(d) \ge 2$ satisfying*

$$L = o\left(\sqrt{\frac{\log d}{\log\log d}}\right).$$

For each d, the width limit is taken with its corresponding $L(d)$ and sample size fixed.

Proof. Here $M = d^{1/(L-1)} \to \infty$. The halfspace construction has $m \le Cn\log(M+1) = o(M) \le o(d)$ uniformly in $L \ge 2$ at the specified sample sizes. Its first-feature entropy bound and amplification are also uniform. The rate ratio is therefore at most

$$CL\, d^{-\eta/[L(L-1)]}[\log(d+1)]^{1/L}$$

with universal C. Its logarithm tends to minus infinity under the stated depth assumption. The mixture hypothesis and the training success probability also hold uniformly, so the same rate-gap argument applies. □

25 The exact first-order scalar entrance cost

Theorem 25.1 (Small-variance scalar expansion)**.** *There are universal constants $q_0, C > 0$ such that*

$$|V(q) - (V_* - 2q)| \le Cq^2 \qquad (0 < q \le q_0).$$

In particular, $\lim_{q\downarrow 0}(V_* - V(q))/q = 2$.

Proof. We first construct a nearly optimal small-state bridge. At state q, replace the reference Gaussian by $\mathcal{N}(0, (1+4q)q)$. Since

$$F(u) = u - 2u^2 + O(u^3),$$

its new state q' and entropy cost D satisfy

$$q' = q + 2q^2 + O(q^3), \qquad D = \tfrac{1}{2}[4q - \log(1+4q)] = 4q^2 + O(q^3).$$

For sufficiently small q, therefore,

$$q' - q \ge q^2, \qquad D \le (2 + Cq)(q' - q).$$

Starting at any $0 < a < b$ in this small interval, take full steps while they remain below b. The states reach the final interval in finitely many steps, since every increment is at least a^2. The full-step costs sum to at most $2(b-a) + Cb^2$. On the last step reduce the variance inflation continuously to hit b exactly; its entropy is at most Cb^2. Thus, after changing C, there is a bridge from a to b of cost

$$2(b-a) + Cb^2. \tag{21}$$

Appending a nearly optimal path from q and letting $a \downarrow 0$ proves

$$V_* - V(q) \le 2q + Cq^2.$$

For the reverse inequality take q still smaller, and start a near-optimal scalar path at $0 < a < q$. Its cost may be assumed less than two, by the uniform upper bound for V. It must cross level q: otherwise the readout alone costs more than $1/(8q) > 2$. Let b be its first state at or above q, and a_- the preceding state. Then $a_- < q$. The crossing satisfies $b < 8q$. Indeed the preactivation second moment is at least b, because $\tanh^2 z \le z^2$. At a ratio $r = b/a_- \ge 8$, Gaussian entropy duality gives a single-step cost at least

$$\tfrac{1}{2}(r - 1 - \log r) \ge \tfrac{1}{2}(8 - 1 - \log 8) > 2,$$

contrary to the total-cost bound.

The scalar exponential potential with coefficient two gives prefix entropy at least $2(b-a)$. The remaining path costs at least $V(b)$. Apply the bridge (21) from q to b, which stays in the small-state interval since $b < 8q$:

$$V(b) \geq V(q) - 2(b-q) - Cb^2.$$

The entire path therefore costs at least

$$V(q) + 2q - 2a - 64Cq^2.$$

Taking the infimum over near-optimal paths and then $a \downarrow 0$ gives $V_* - V(q) \geq 2q - C''q^2$, proving the theorem. □

The coefficient two describes the scalar entrance to the common rare-event trajectory. It does not identify the function-dependent cost of bringing a particular target direction to that scalar regime. Thus it is an ingredient for a possible target-relative expansion, not a characterization of the posterior's learning behavior.

A An exact variational principle at the critical exponent

For completeness we reproduce the finite-domain variational foundation. Here $\phi = \tanh$. For arbitrary fixed depth, the critical prior admits a direct finite-domain large-deviation description. This result requires neither a propagation-of-chaos assumption nor differentiability of the activation. It identifies the functional used in the critical-rate capacity constructions with the actual Gaussian probability rate. Covariance-process large deviations for Gaussian networks have been studied in a more general functional setting by Andreis, Bassetti, and Hirsch [2]. No novelty claim is made for the general large-deviation framework; the finite-domain proof below fixes the precise normalizations and strict-margin convention used here.

Fix $\gamma = 1$, enumerate $\mathcal{X} = \{x^1, \dots, x^m\}$, where $m = 2^d$, and put $r = L - 1$. All limits in this section hold with d and L fixed. Write $E = [-1,1]^m$ and let $\mathcal{P}(E)$ carry its weak topology. For $\nu \in \mathcal{P}(E)$ define

$$K(\nu) = \int_E hh^{\mathsf{T}}\, d\nu(h), \qquad Q_0(a,b) = \frac{\bar{x}^a \cdot \bar{x}^b}{d}.$$

For any positive semidefinite $m \times m$ matrix Q, let

$$\mathsf{T}_Q(u) = \left(\phi((Q^{1/2}u)_a)\right)_{a=1}^m, \qquad \mathsf{M}_Q = (\mathsf{T}_Q)_{\#}\mathcal{N}(0, I_m).$$

Here $T_{\#}\rho$ is the distribution of $T(U)$ when U has distribution ρ, and $Q^{1/2}$ is the positive semidefinite square root. Relative entropy is

$$D(\nu\|\rho) = \begin{cases} \displaystyle\int \log \frac{d\nu}{d\rho}\, d\nu, & \nu \ll \rho, \\ +\infty, & \text{otherwise.} \end{cases}$$

Finally define the Gaussian energy

$$\mathcal{G}_Q(z) = \begin{cases} \frac{1}{2} z^{\mathsf{T}} Q^{+} z, & z \in \operatorname{range}(Q), \\ +\infty, & z \notin \operatorname{range}(Q), \end{cases}$$

where Q^+ is the Moore–Penrose inverse.

Theorem A.1 (Critical finite-domain large deviations)**.** *Let $Z_N = (z_w(x^a))_{a=1}^m$ under the unconditioned prior $\mu_{N,1}$. The sequence Z_N satisfies a large-deviation principle with speed N and good rate function*

$$I(z) = \inf_{\nu_1,\dots,\nu_r \in \mathcal{P}(E)} \left\{ D(\nu_1 \| \mathsf{M}_{Q_0}) + \sum_{s=2}^{r} D(\nu_s \| \mathsf{M}_{K(\nu_{s-1})}) + \mathcal{G}_{K(\nu_r)}(z) \right\}. \tag{22}$$

In particular, for every open $O \subseteq \mathbb{R}^m$ and closed $F \subseteq \mathbb{R}^m$,

$$-\inf_{z \in O} I(z) \le \liminf_{N\to\infty} \frac{1}{N} \log \mathbb{P}(Z_N \in O), \qquad \limsup_{N\to\infty} \frac{1}{N} \log \mathbb{P}(Z_N \in F) \le -\inf_{z \in F} I(z).$$

The infimum in (22) is attained whenever it is finite. No covariance matrix in the statement is required to be nonsingular.

Proof. We give an exact representation of the finite-width law and then apply Sanov's theorem and continuous contraction [3].

Let $\Gamma_m = \mathcal{N}(0, I_m)$. Independently for each $s = 1, \dots, r$, let $U_{s,1}, \dots, U_{s,N}$ have law Γ_m, and set

$$\eta_{s,N} = \frac{1}{N} \sum_{i=1}^{N} \delta_{U_{s,i}}.$$

Also let $V_N = G/\sqrt{N}$, where $G \sim \Gamma_m$ is independent of these empirical measures. Recursively define

$$\nu_{1,N} = (\mathsf{T}_{Q_0})_{\#}\eta_{1,N}, \qquad \nu_{s,N} = (\mathsf{T}_{K(\nu_{s-1,N})})_{\#}\eta_{s,N} \quad (2 \le s \le r).$$

These are exactly the joint laws of the network's empirical hidden-activation vectors on $\mathcal{X}$. Indeed, first-layer preactivation vectors are independent Gaussians with covariance Q_0. Conditional on the preceding layer, each subsequent row of weights gives an independent Gaussian vector with covariance

$$\frac{1}{N} \sum_{i=1}^{N} \big(h_i^{s-1}(x^a) h_i^{s-1}(x^b)\big)_{a,b} = K(\nu_{s-1,N}).$$

The conditional distribution depends on the previous layer only through its empirical measure. Finally, readout weights have variance N^{-2}, so

$$Z_N \overset{\text{law}}{=} K(\nu_{r,N})^{1/2} V_N. \tag{23}$$

This representation is exact for every N.

Sanov's theorem on $\mathcal{P}(\mathbb{R}^m)$ gives the good rate $D(\eta \| \Gamma_m)$ for each $\eta_{s,N}$. The elementary Gaussian large-deviation principle gives rate $\|v\|_2^2/2$ for V_N. Independence and the finite-product principle therefore give the good rate

$$\sum_{s=1}^{r} D(\eta_s \| \Gamma_m) + \frac{1}{2}\|v\|_2^2$$

for $(\eta_{1,N}, \dots, \eta_{r,N}, V_N)$.

The recursive map from these variables to $(\nu_1, \dots, \nu_r, z)$ is continuous. To verify the only potentially delicate point, suppose $Q_j \to Q$ and $\eta_j \Rightarrow \eta$ weakly. For every continuous $b : E \to \mathbb{R}$, the functions $b \circ \mathsf{T}_{Q_j}$ converge uniformly on each compact subset of $\mathbb{R}^m$ to $b \circ \mathsf{T}_Q$, since matrix square

roots are continuous and ϕ is continuous. Their absolute values are uniformly bounded. Tightness of a weakly convergent sequence of probability measures then gives

$$\int b \circ \mathsf{T}_{Q_j}\, d\eta_j \longrightarrow \int b \circ \mathsf{T}_Q\, d\eta.$$

Also $\nu \mapsto K(\nu)$ is continuous because every coordinate $h_a h_b$ is continuous and bounded on E. This proves continuity of the whole recursive map, including (23).

Continuous contraction now gives the rate by minimizing over its preimages. For a measurable map T and a reference probability ρ, the entropy contraction identity is

$$\inf_{\eta: T_\# \eta = \nu} D(\eta \| \rho) = D(\nu \| T_\# \rho). \tag{24}$$

The lower bound is data processing. When the right side is finite and $q = d\nu/d(T_\#\rho)$, the measure $d\eta = q(T(u))\, d\rho(u)$ attains equality. If the right side is infinite, data processing already proves the identity in the extended real sense. Apply (24) successively with $T = \mathsf{T}_{Q_0}$ and $T = \mathsf{T}_{K(\nu_{s-1})}$. The last minimization is

$$\inf_{v: Q^{1/2} v = z} \frac{1}{2}\|v\|_2^2 = \mathcal{G}_Q(z),$$

including singular Q. This proves (22).

The contraction theorem supplies a good rate and exponential tightness. For completeness, the displayed minimization is also directly well behaved: relative entropy is jointly lower semicontinuous, the map $\nu \mapsto \mathsf{M}_{K(\nu)}$ is weakly continuous, and

$$\mathcal{G}_Q(z) = \sup_{t \in \mathbb{R}^m} \left\{ t^\mathsf{T} z - \frac{1}{2} t^\mathsf{T} Q t \right\}$$

is jointly lower semicontinuous in (Q, z). Since $\mathcal{P}(E)^r$ is compact, every finite infimum in (22) is attained. Moreover $K(\nu) \preceq m I_m$, so $I(z) \geq \|z\|_2^2/(2m)$. □

Corollary A.2 (Full-margin masses and posterior rates)**.** *For a Boolean function f, regarded as a vector in $\{-1,1\}^m$, set*

$$J(f) = \inf_{z:\ f_a z_a \geq 1/2\ \forall a} I(z).$$

Then the strict-margin event used throughout this paper has the exact rate

$$\lim_{N\to\infty} -\frac{1}{N} \log \mu_{N,1}(A_{N,f}) = J(f). \tag{25}$$

For a fixed noiseless training set S, let $\mathcal{B}_S$ be the Boolean functions agreeing with its labels. If $J_S = \min_{g \in \mathcal{B}_S} J(g) < \infty$, the actual globally conditioned posterior satisfies, for every $f \in \mathcal{B}_S$,

$$\lim_{N\to\infty} -\frac{1}{N} \log \Pi_{N,1}(f \mid S) = J(f) - J_S.$$

In particular, posterior mass concentrates on the minimizers of J within $\mathcal{B}_S$. These logarithmic asymptotics do not determine how mass is divided between tied minimizers.

Proof. For $c \geq 1$, retaining any feature laws in (22) and scaling only the output gives

$$I(cz) \leq c^2 I(z),$$

because the entropy terms are nonnegative and the Gaussian energy scales quadratically. If $f_a z_a \geq 1/2$ for all a, then cz has strict margin for every $c > 1$. Letting $c \downarrow 1$ shows that the infimum of I over the open strict-margin orthant equals its infimum over the closed orthant. The lower and upper large-deviation bounds therefore agree, proving (25).

Conditioning the prior on the global margin event and subsequently on the labels gives exactly

$$\Pi_{N,1}(f \mid S) = \frac{\mu_{N,1}(A_{N,f})}{\sum_{g \in \mathcal{B}_S} \mu_{N,1}(A_{N,g})}.$$

For fixed d this is a finite sum, whose exponential rate is the smallest $J(g)$ among its terms. This proves the ratio formula and concentration claim. □

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